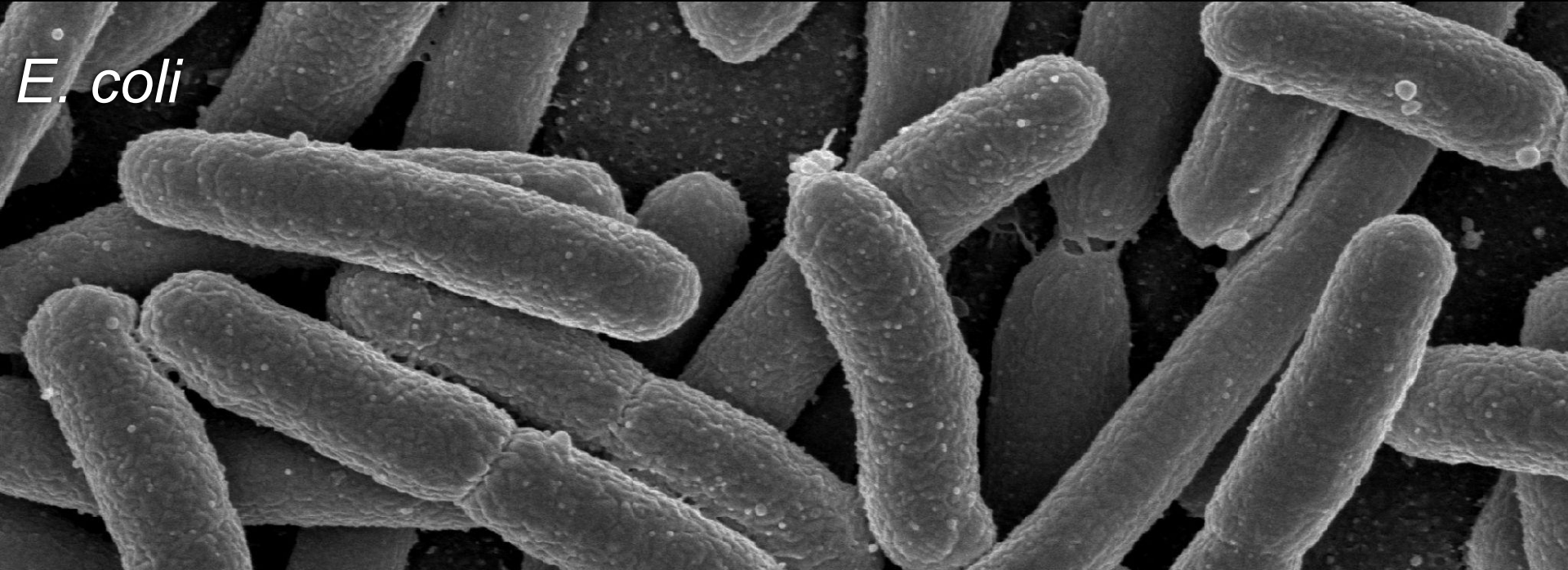


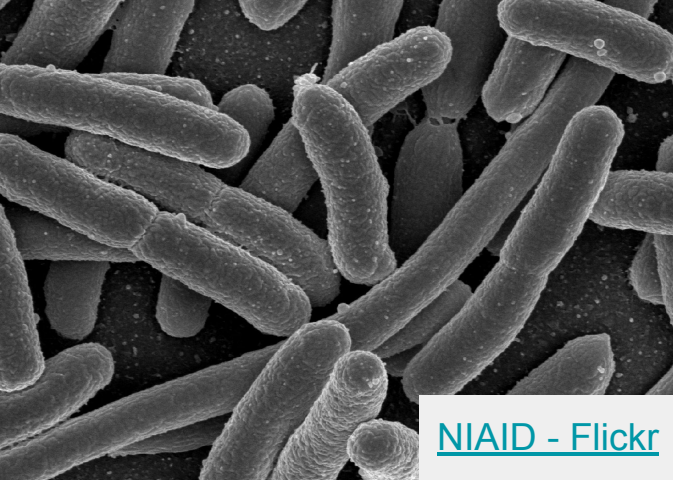


E. coli

Overview of evolutionary computing
Dr. Austin Ferguson (he/him)
2026-09-28 ferguaus@gvsu.edu



[Niko photos - Unsplash](#)

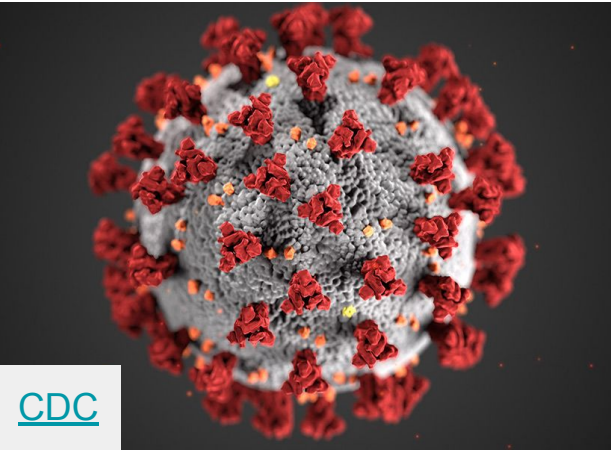


[NIAID - Flickr](#)

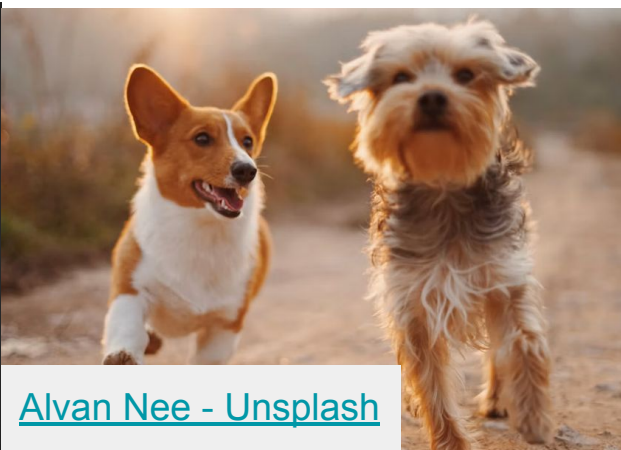


[Anna Roberts - Unsplash](#)

What is evolution?



[CDC](#)



[Alvan Nee - Unsplash](#)



[MD_JERRY - Unsplash](#)

Clarifying terms

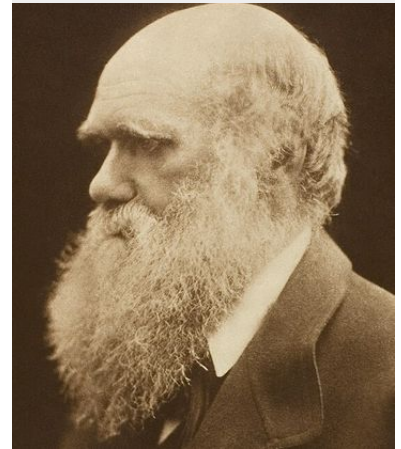
Evolution - Genetic change in a population over time

Clarifying terms

Evolution - Genetic change in a population over time

Natural selection - Evolution steered by natural causes

- e.g., rising temperatures selecting for higher heat tolerance



Clarifying terms

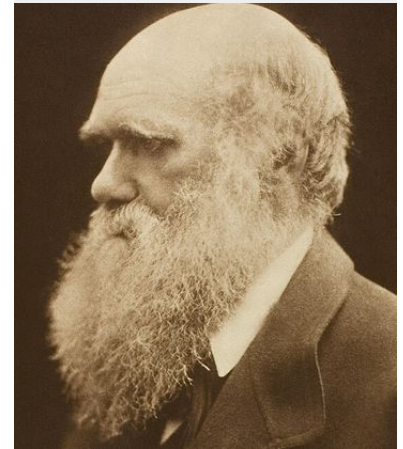
Evolution - Genetic change in a population over time

Natural selection - Evolution steered by natural causes

- e.g., rising temperatures selecting for higher heat tolerance

Artificial selection - Evolution driven by us!

- e.g., breeding dogs to make them smaller



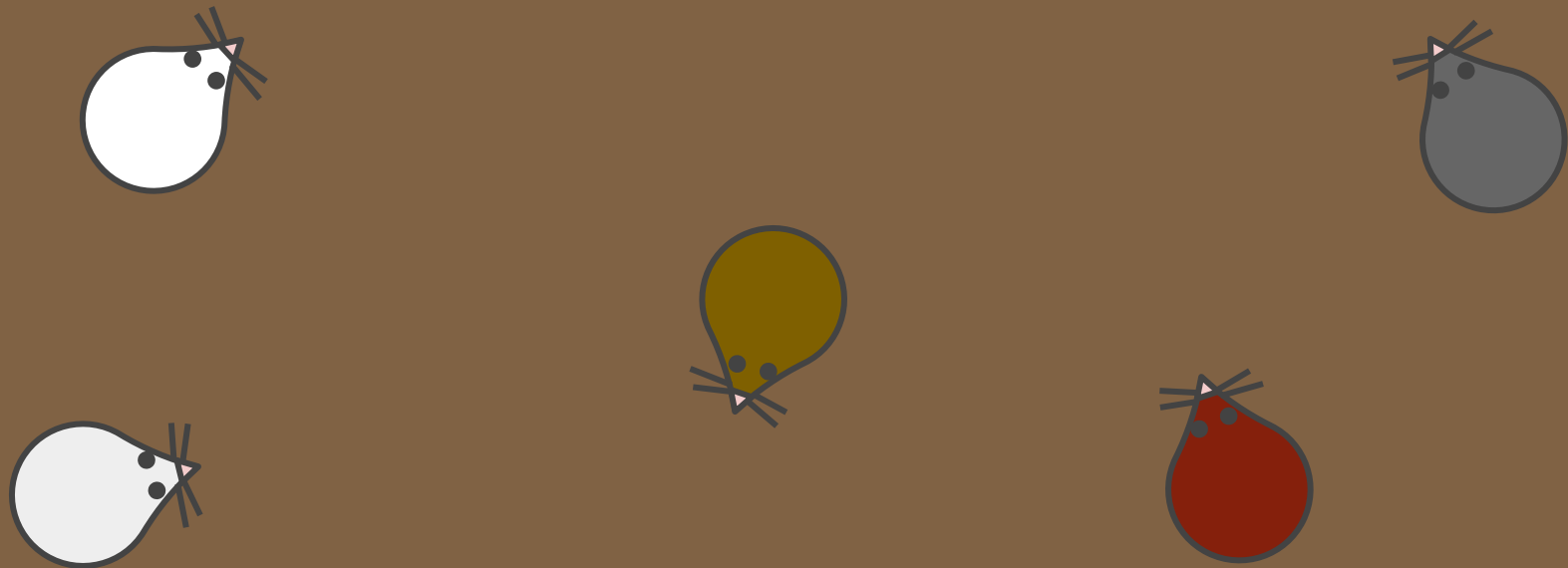
Bridging CS and evolution

What does evolution actually do?

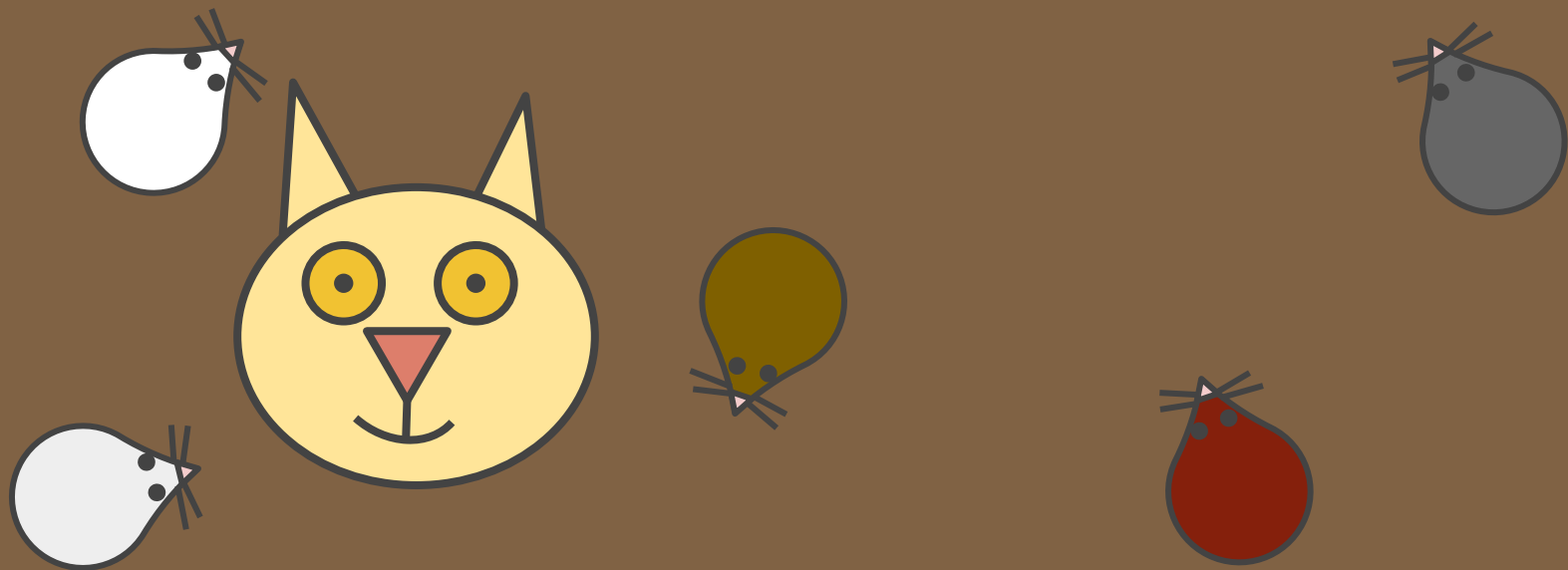
Evolution example



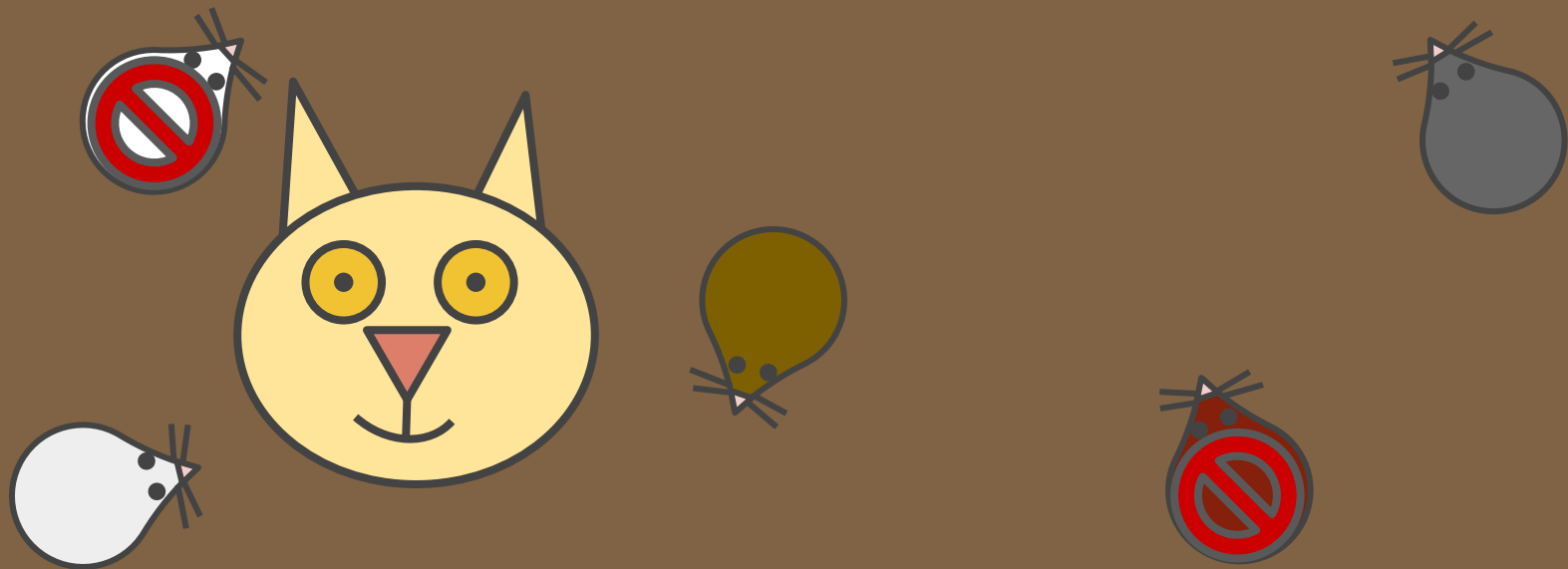
Evolution example



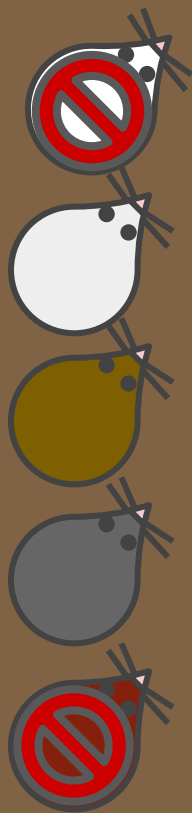
Evolution example



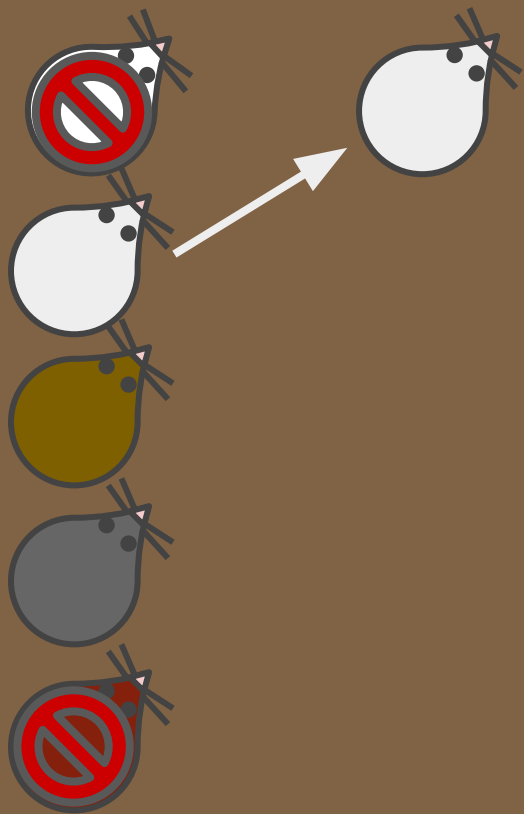
Evolution example



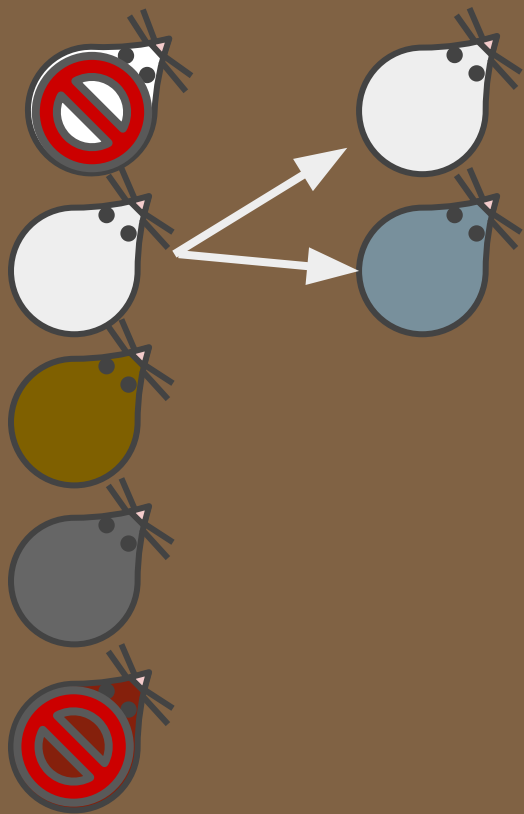
Evolution example



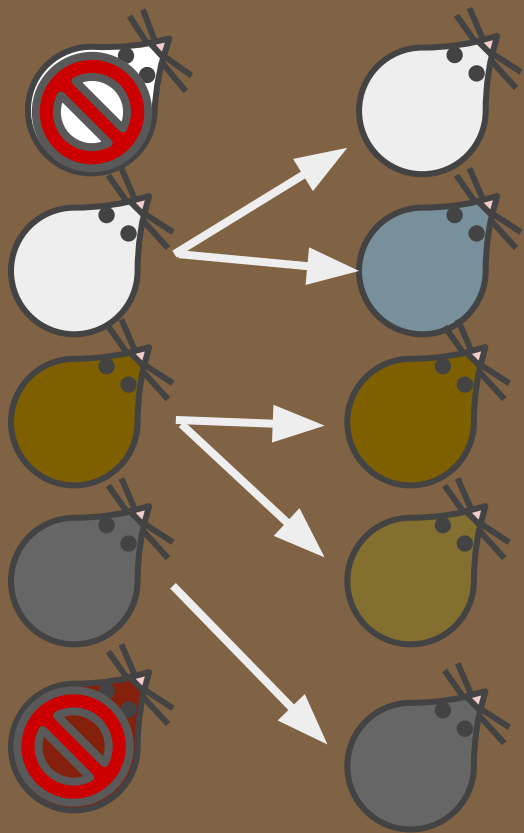
Evolution example



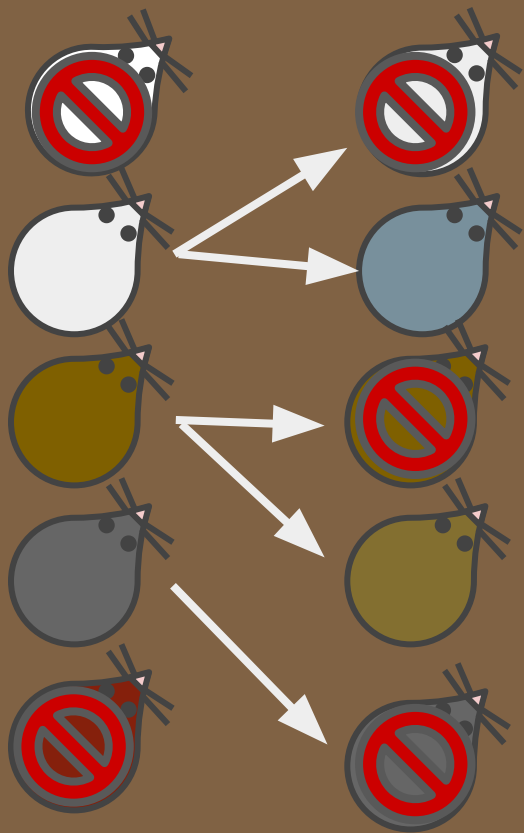
Evolution example



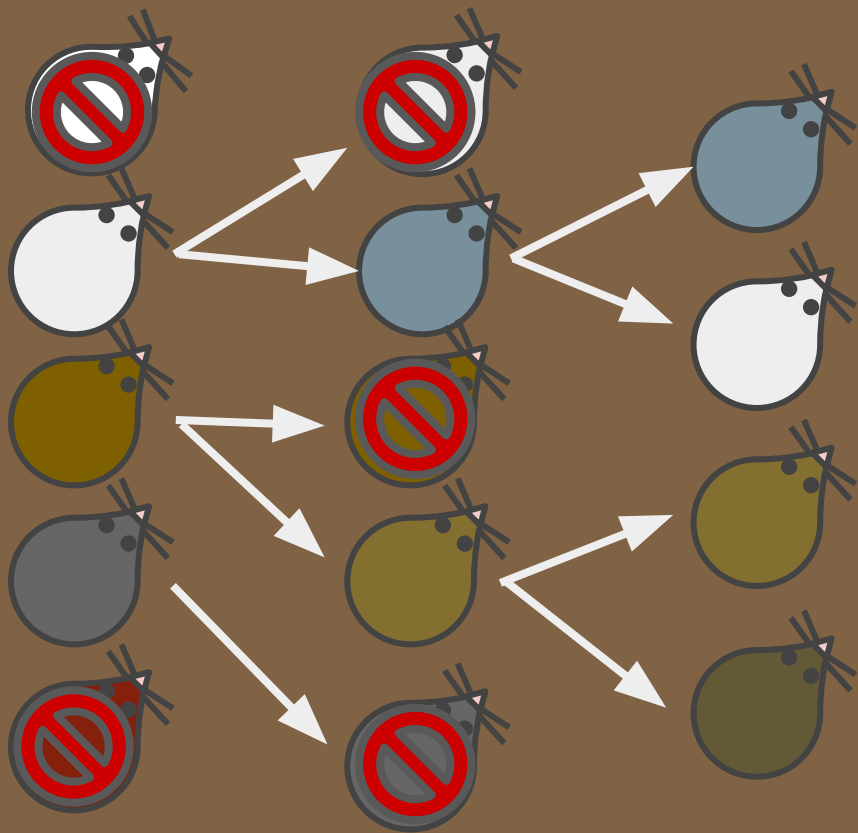
Evolution example



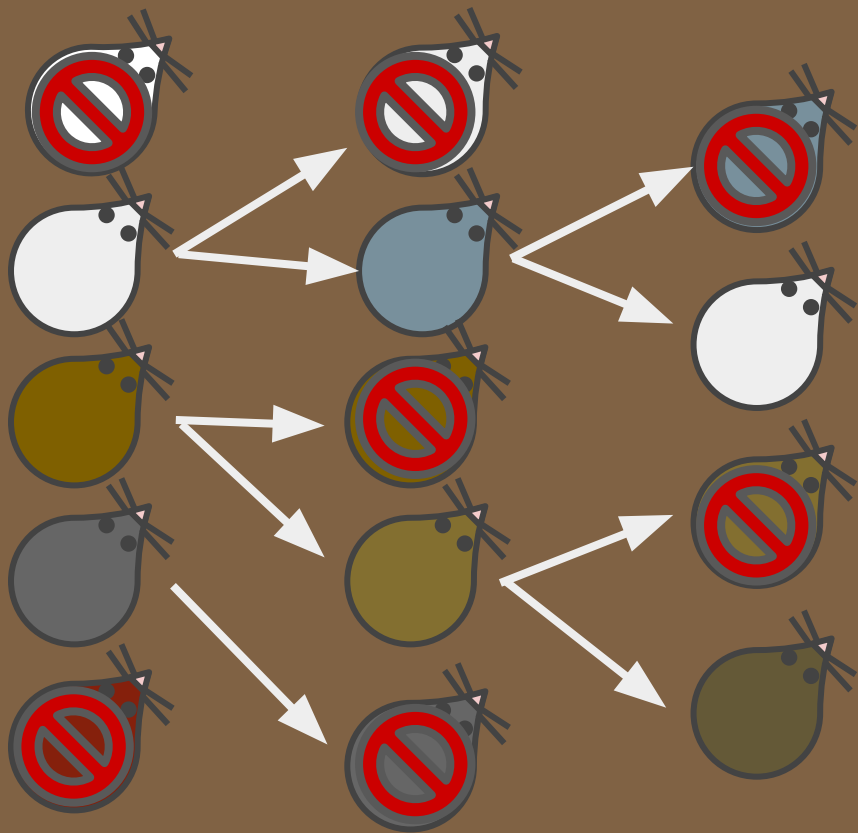
Evolution example



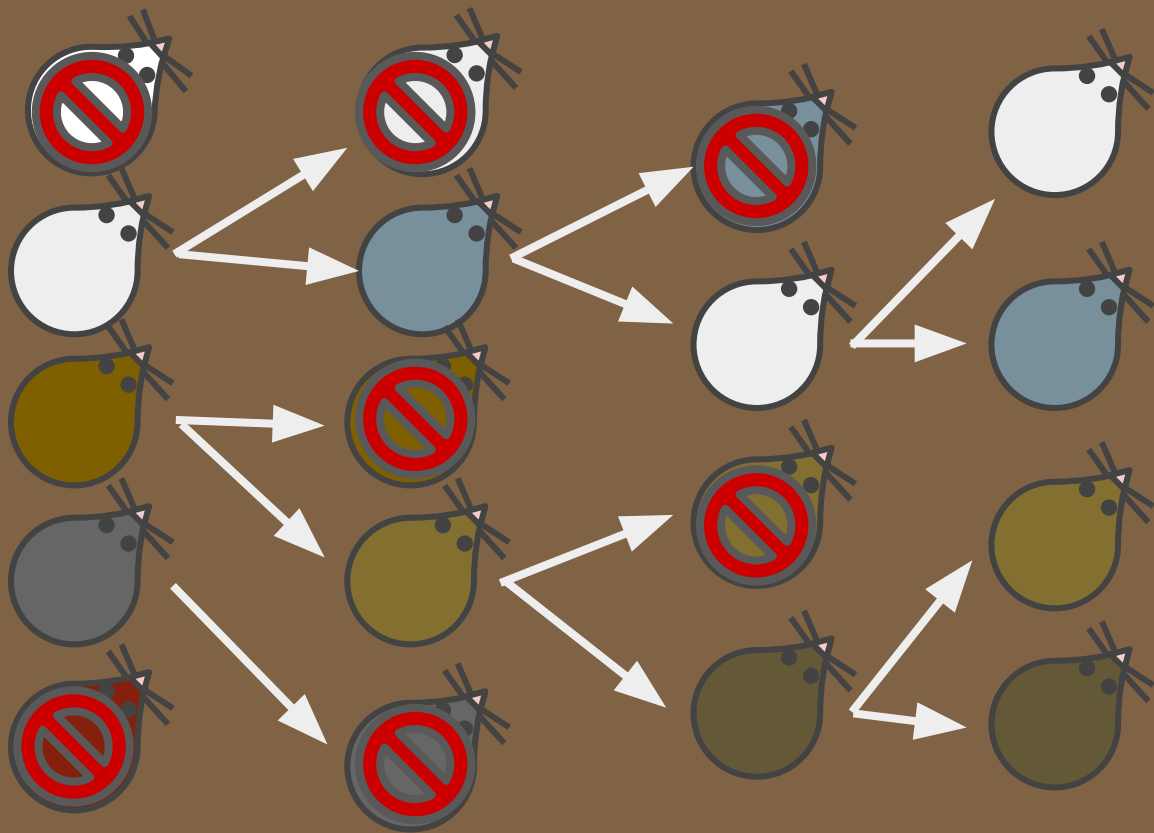
Evolution example



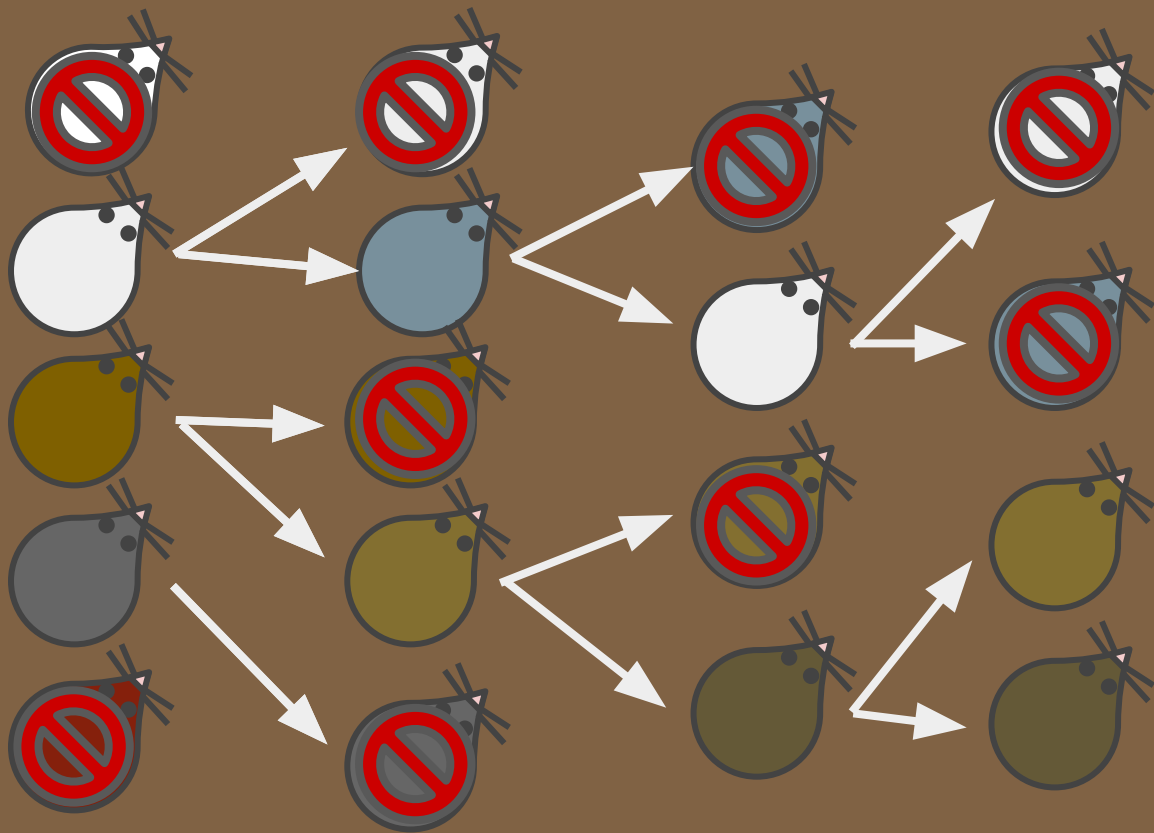
Evolution example



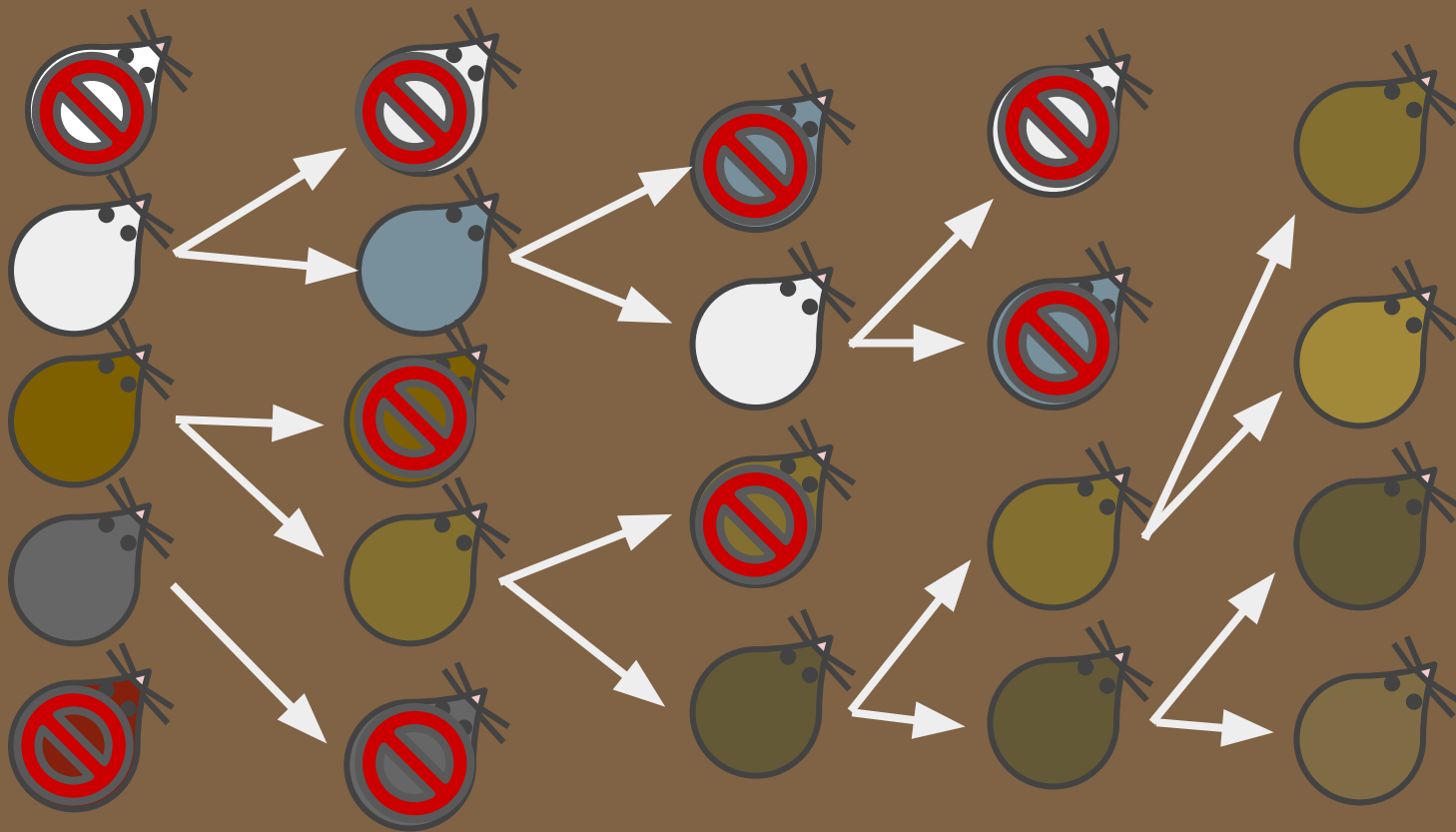
Evolution example



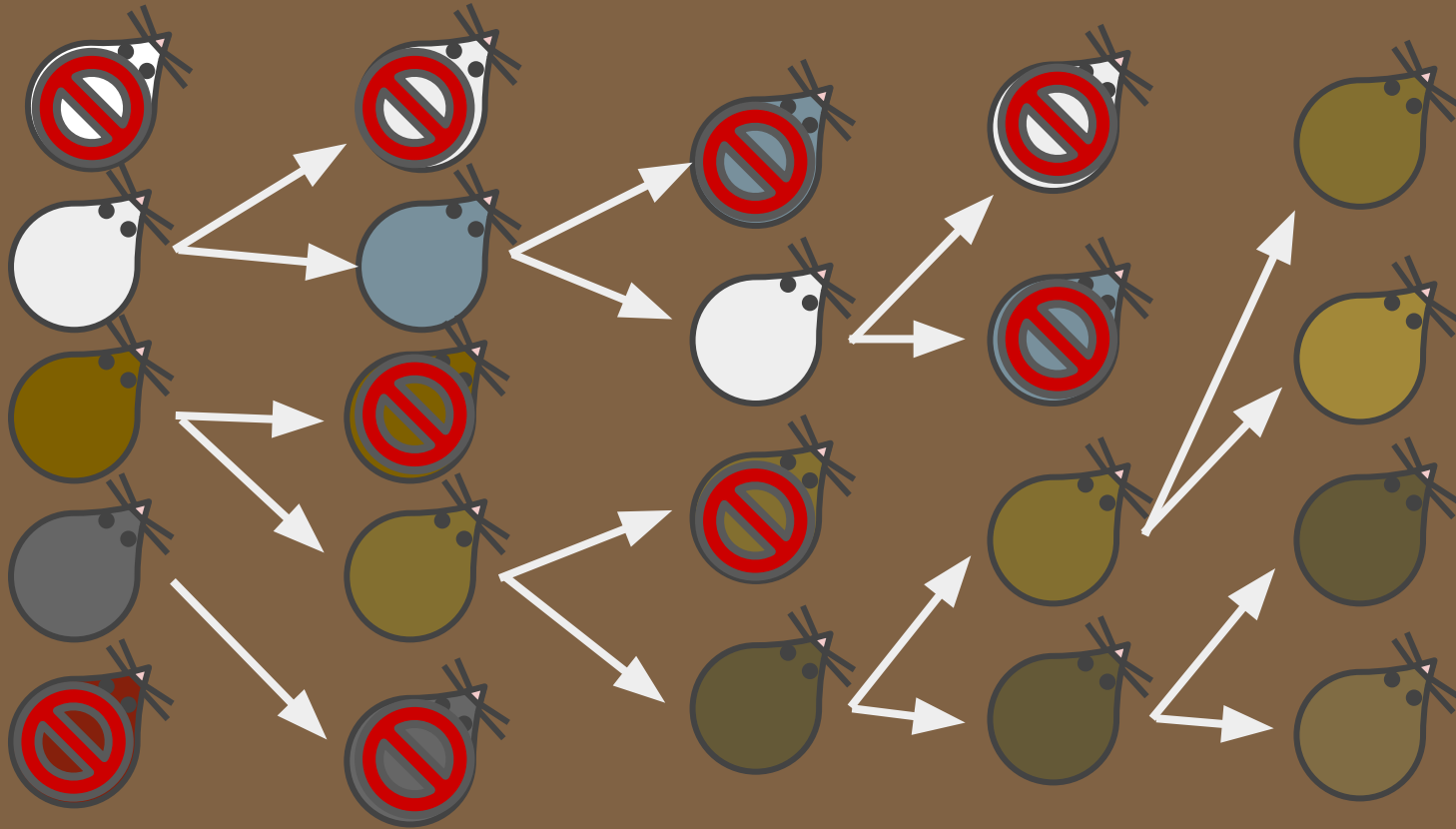
Evolution example



Evolution example

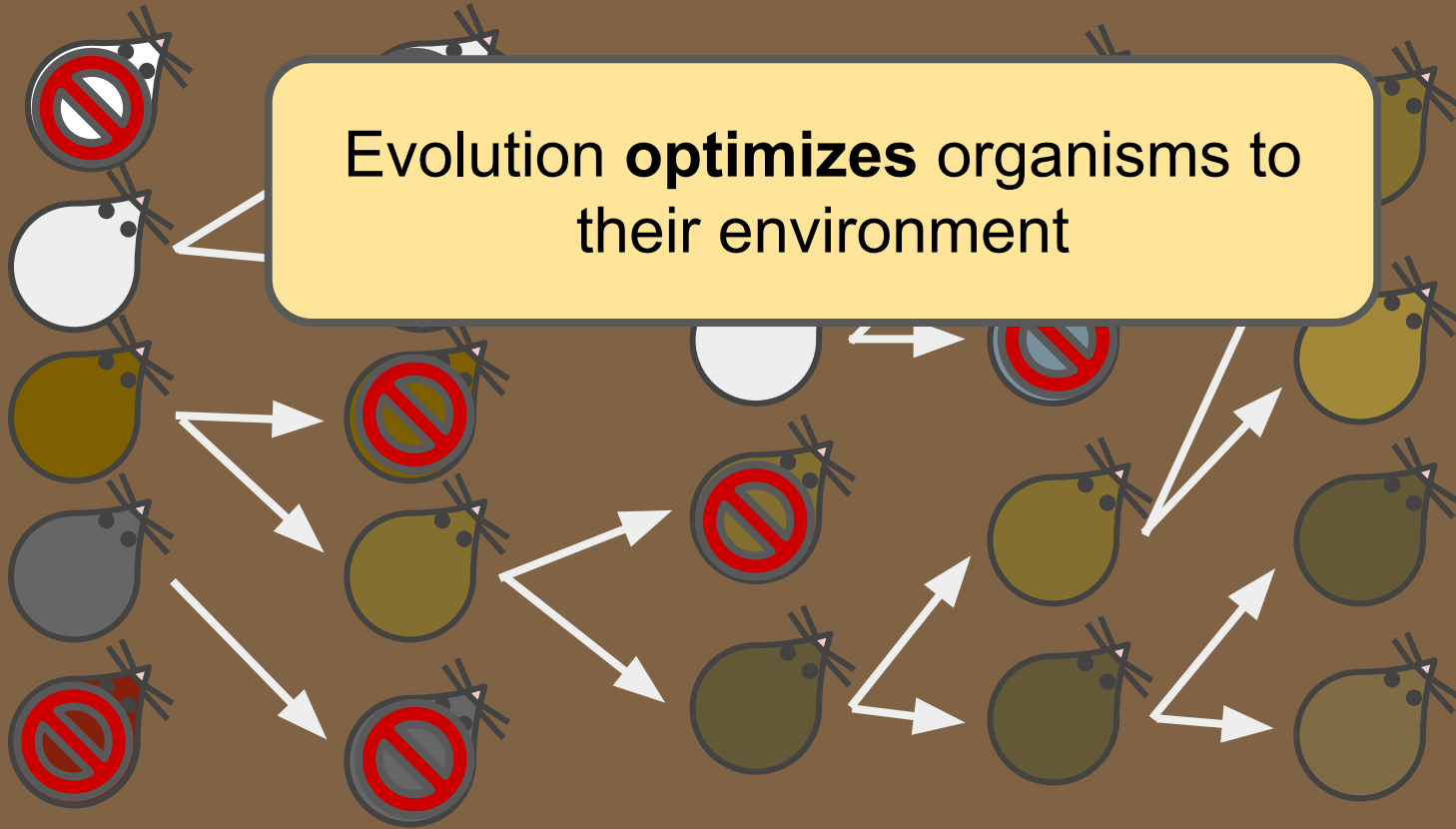


What has evolution done to these organisms?

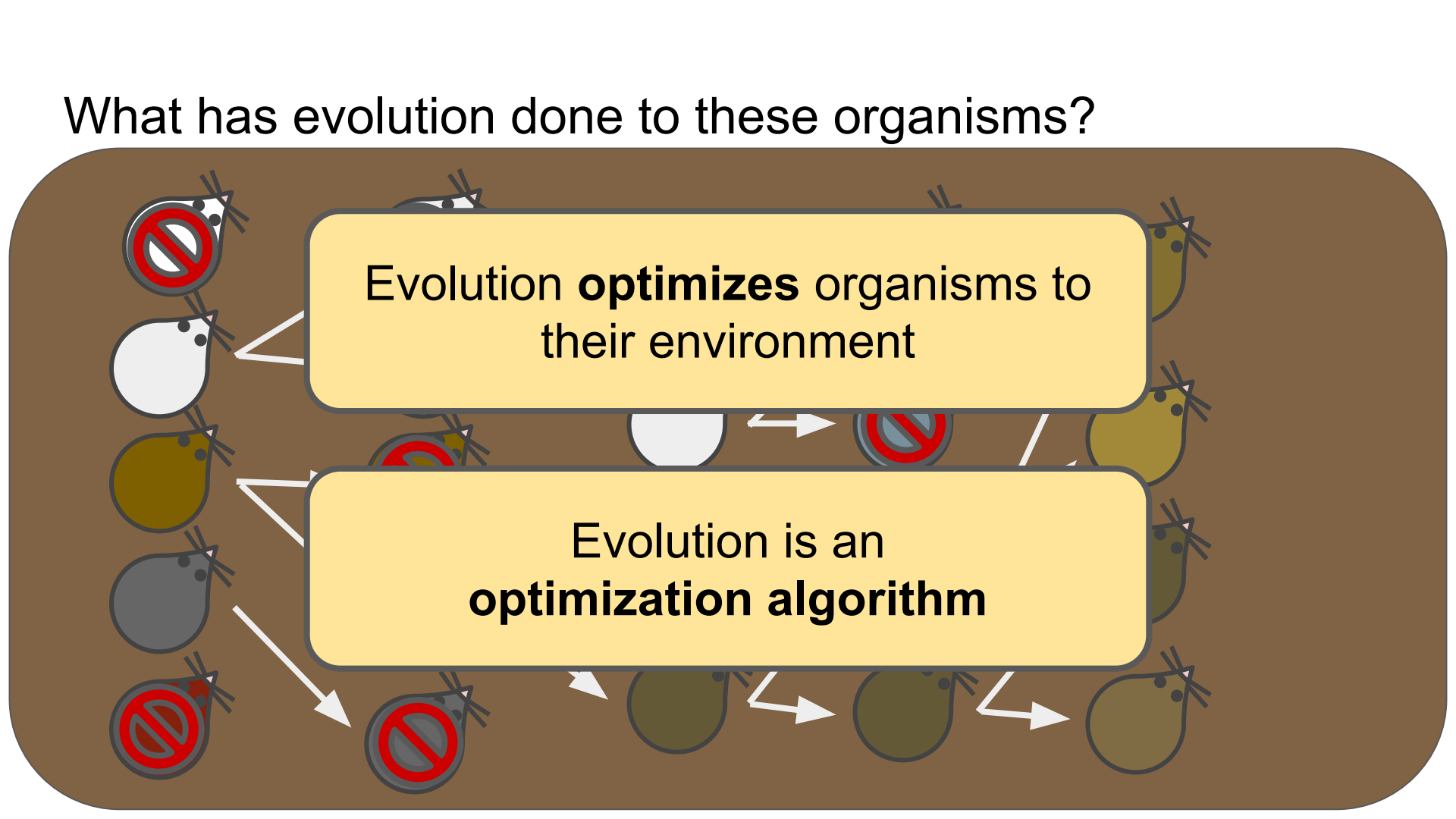


What has evolution done to these organisms?

Evolution **optimizes** organisms to
their environment



What has evolution done to these organisms?



The diagram illustrates evolution as an optimization algorithm using a population of stylized organisms. The organisms are represented as simple shapes with three eyes and a tail. They are colored white, olive green, and grey. Some organisms have a red circle with a diagonal slash over their head, indicating they are 'fittest' or 'selected'. White arrows show the flow of the process: from a white organism to a yellow box, then to an olive green organism, then to another yellow box, and finally to a grey organism. This sequence represents the iterative process of selection and adaptation. The background is a solid brown color.

Evolution **optimizes** organisms to
their environment

Evolution is an
optimization algorithm

Evolution in action: the Mega-Plate

[\(youtube link\)](#)



Bringing it back to CS

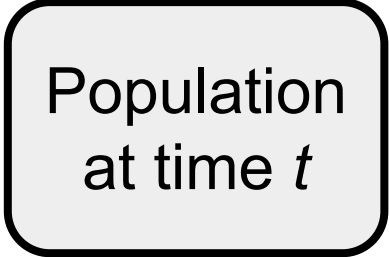
Evolution is an optimization algorithm

So how do we leverage it as one?

What are the required pieces for evolution to occur?

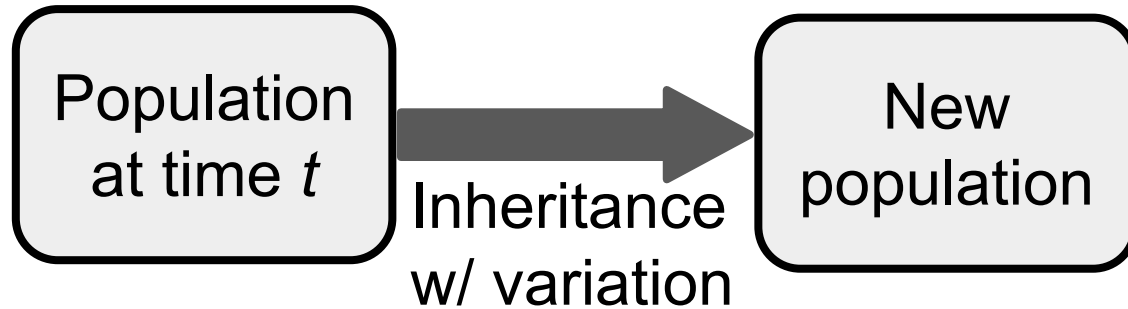
1. **Inheritance** - Offspring should be *similar* to their parents
2. **Variation** - Offspring cannot *always* be identical to their parents
3. **Differences in fitness** - Some organisms are better at surviving & reproducing

Evolution as a process



Population
at time t

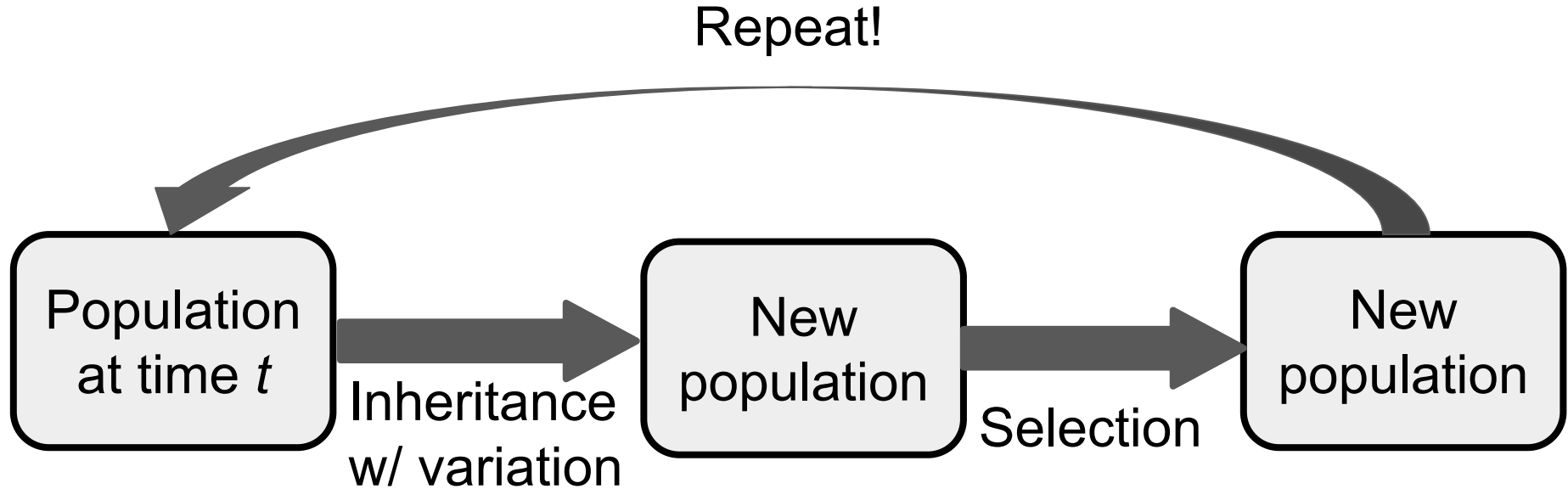
Evolution as a process



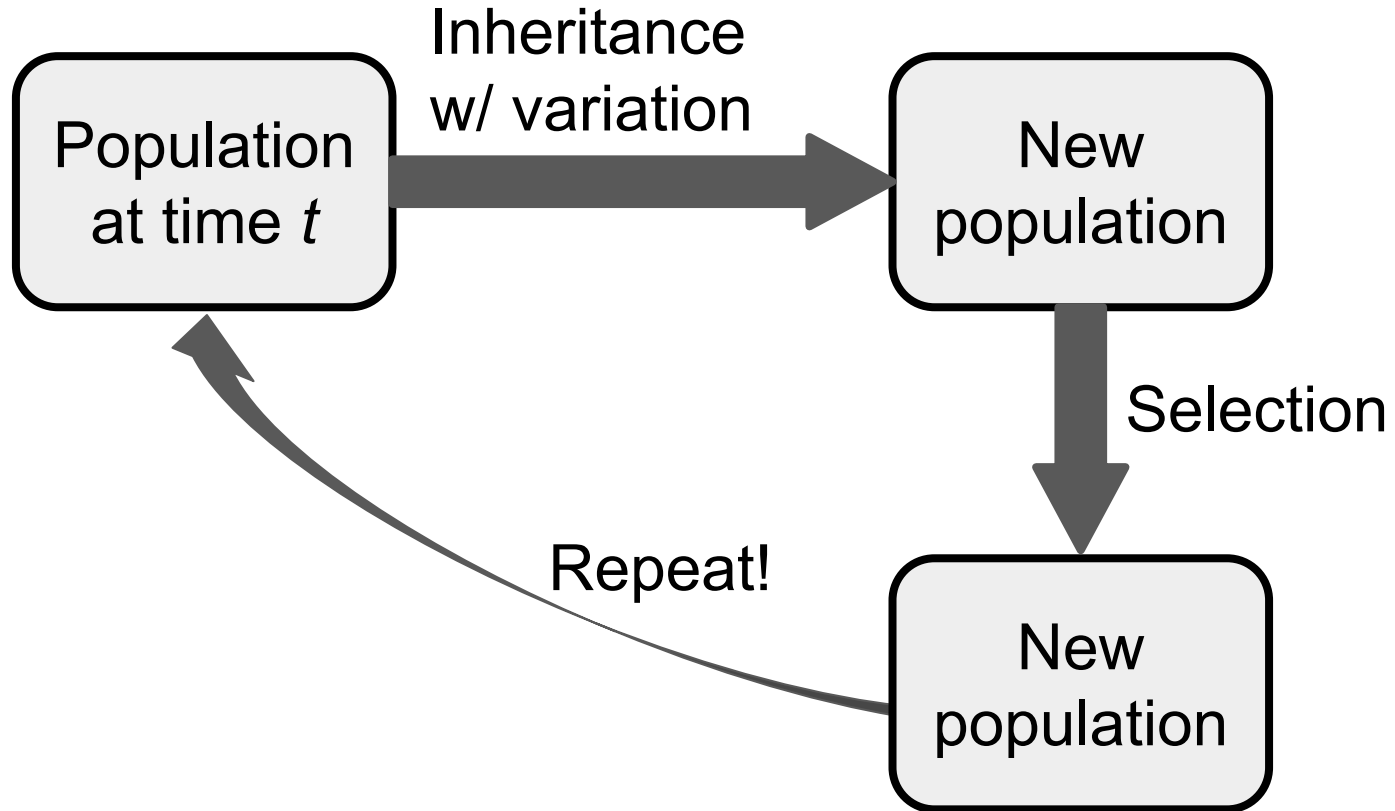
Evolution as a process



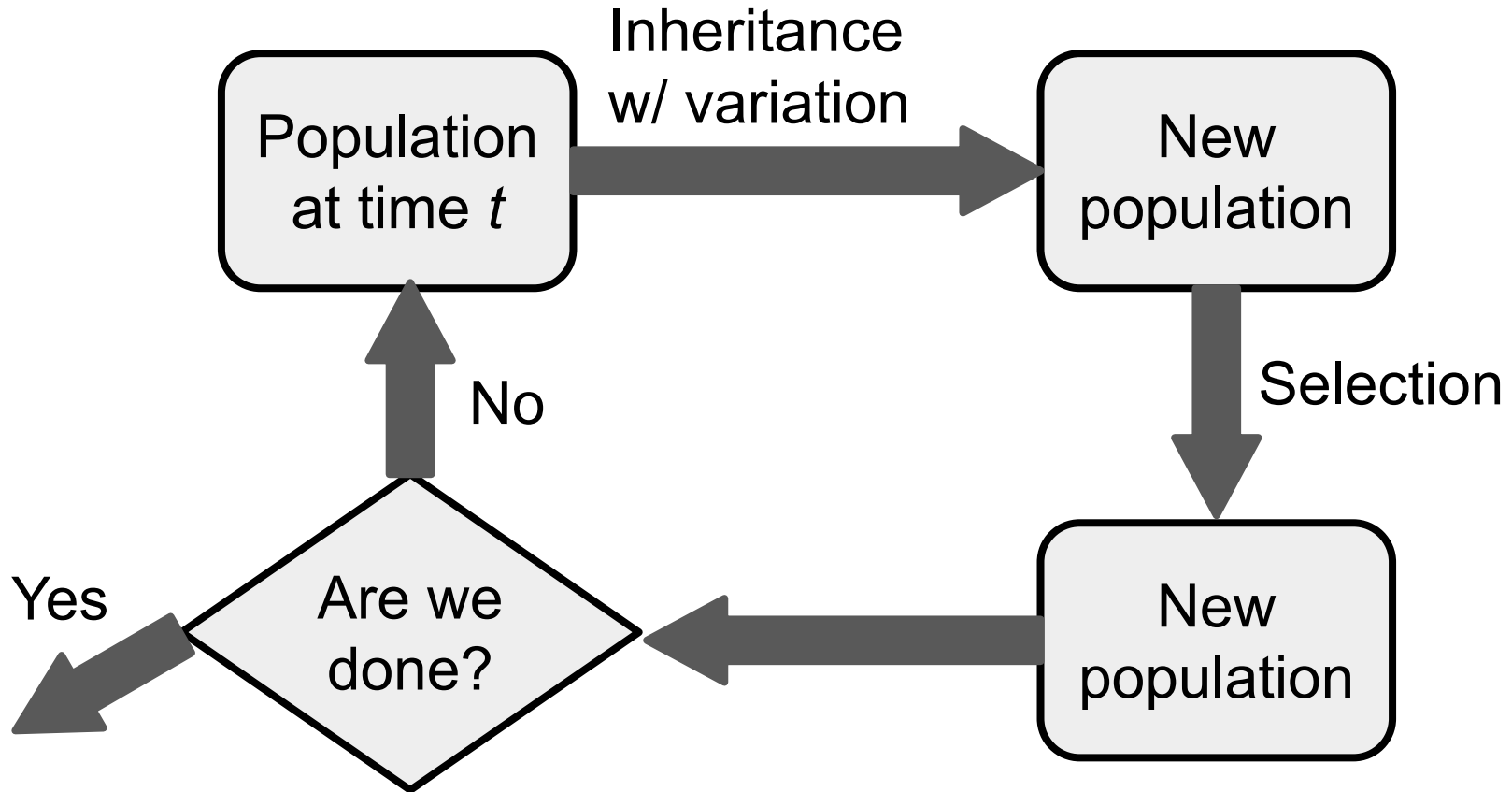
Evolution as a process



Evolution as a process



Evolution as a process

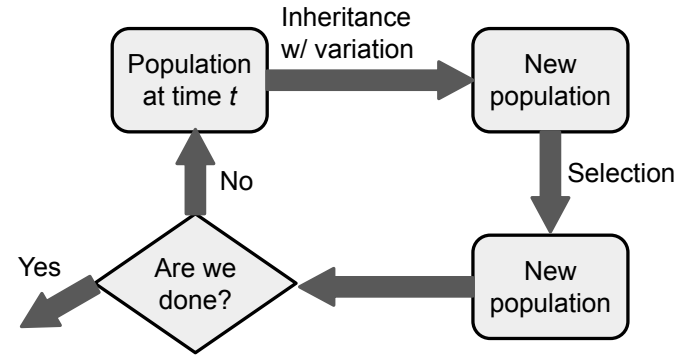


Working through a simple example

Let's evolve the string "CIS 678 - Machine Learning"

Questions for us to answer:

1. What is an organism / solution?
2. How big is our population?
3. How do we create variance?
4. How do we select which organisms reproduce?
5. When should we stop?



String evolution: demo

Link to GitHub repo:

https://github.com/FergusonAJ/ec_demo_cis_678/tree/main

Working through a more interesting example

Imagine you're starting a company and you have a list of tasks you need covered (payroll, sales, software development, etc).

You want to hire the minimum number of people to cover every task.

Given a list of candidates and their skills, who do you hire?

(Note: this is probably a bad way to start a company...)

Set cover problem

Person
A
B
C
D
E
F
G
H

Set cover problem

Person	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8
A								
B								
C								
D								
E								
F								
G								
H								

Set cover problem

Person	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8
A	X	X						
B								
C								
D								
E								
F								
G								
H								

Set cover problem

Person	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8
A	X	X						
B		X			X			X
C								
D								
E								
F								
G								
H								

Set cover problem

Person	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8
A	X	X						
B		X			X			X
C			X	X				
D			X					X
E		X			X			
F	X	X			X			
G					X			X
H				X		X	X	

Set cover problem

Questions for us to answer:

1. What is an organism / solution?
2. How big is our population?
3. How do we create variance?
4. How do we select which organisms reproduce?
5. When should we stop?

	1	2	3	4	5	6	7	8
A	X	X						
B		X			X			X
C			X	X				
D			X					X
E		X			X			
F	X	X			X			
G					X			X
H				X		X	X	



Clarifying Terms

Terms

The field of using evolution as an optimization algorithm for computational problems is [evolutionary computation](#)

The algorithms themselves are called [evolutionary algorithms](#)

"Basic" evo. algorithms that are evolving bitstrings/strings are called [genetic algorithms](#)

Beyond genetic algorithms, the field is split into many subfields:

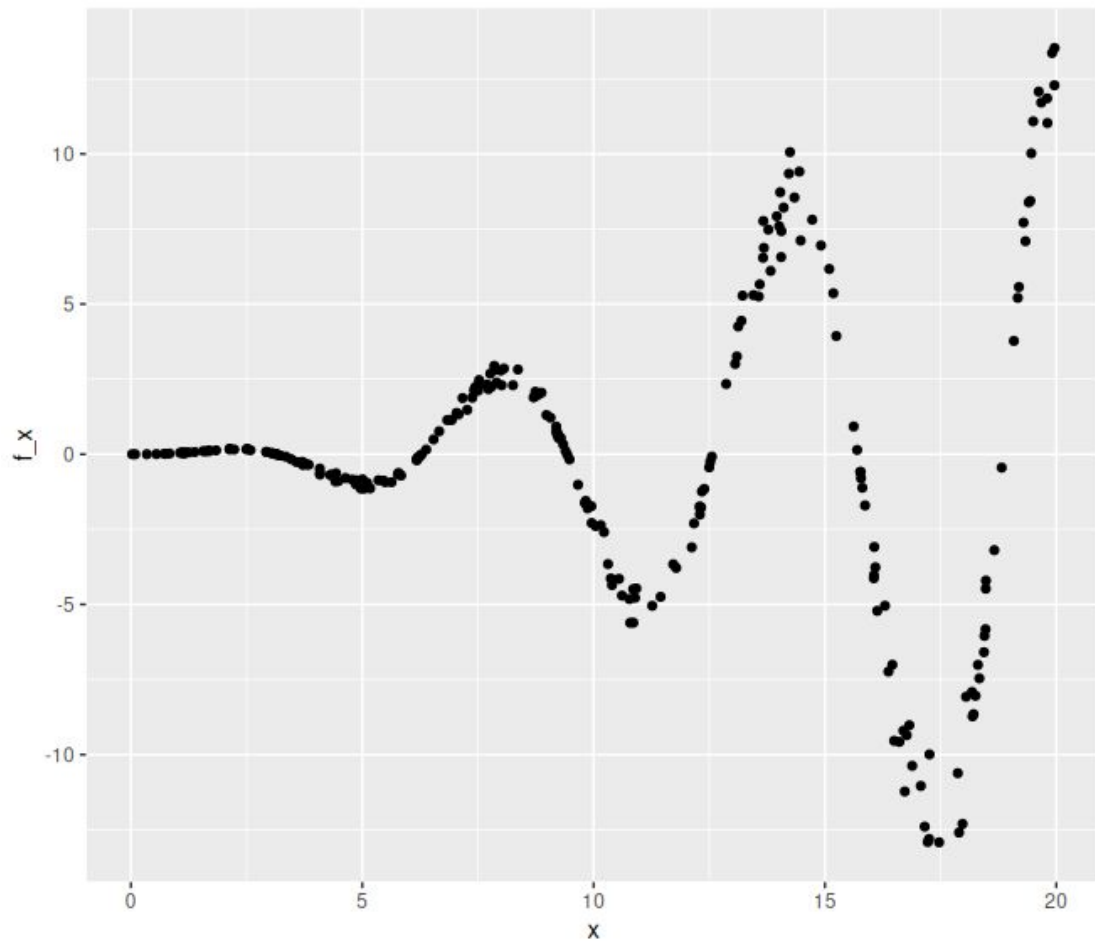
Evolutionary strategies, [genetic programming](#), evolutionary programming, [particle swarm optimization](#), [differential evolution](#), grammatical evolution, [neuroevolution](#), [quality diversity algorithms](#), etc.

Why so many fields?

1. Academics like to "stake their claim"
2. Different problems can involve substantially different algorithms

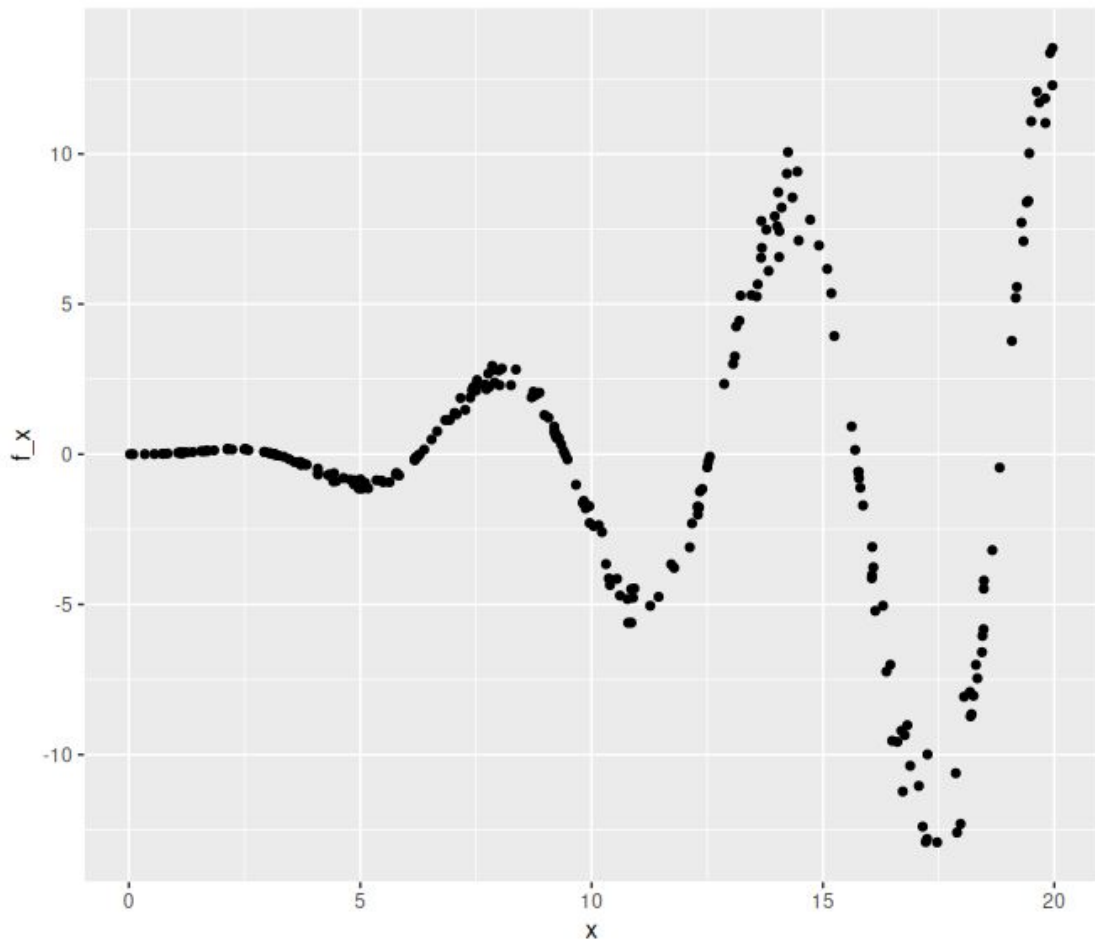
Imagine we have this data:

What can we do with it?



Traditional regression can fit
a given model to the data

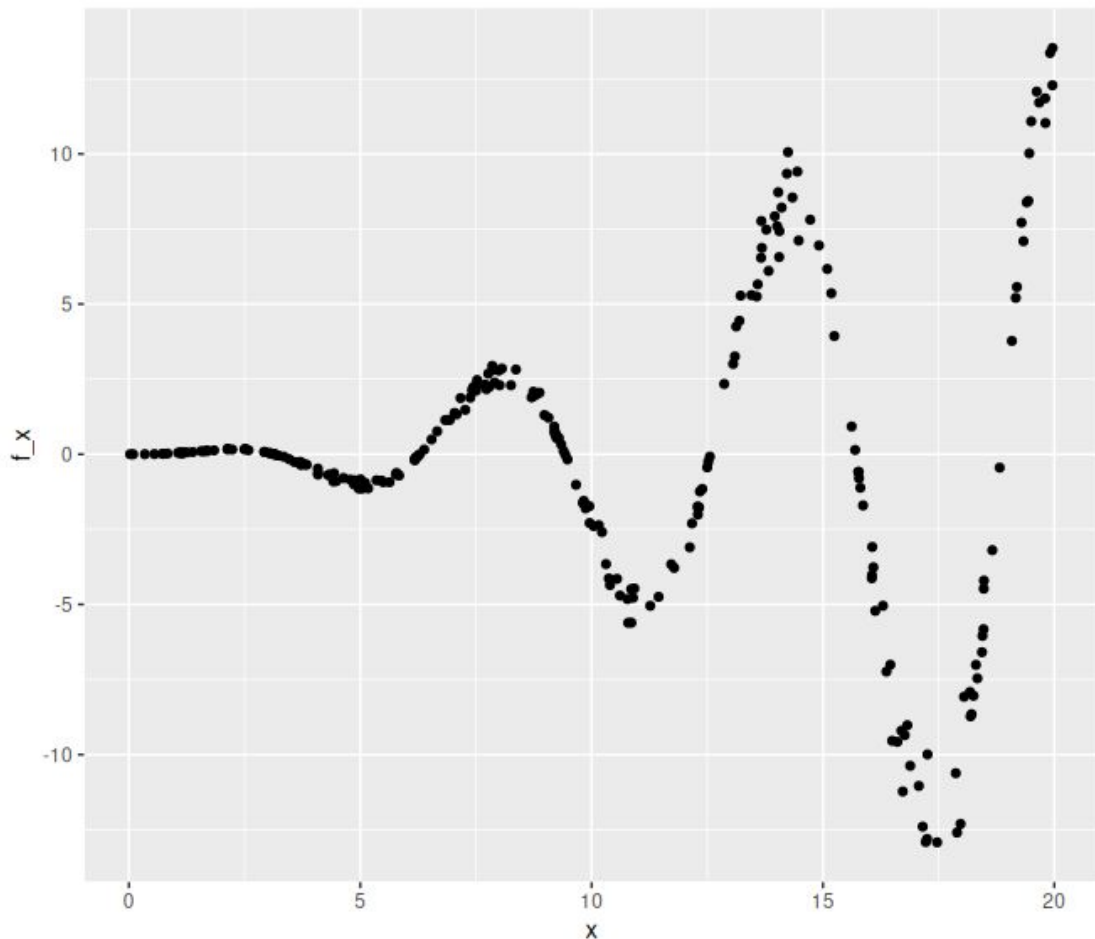
But what if we don't know
the model?



Traditional regression can fit
a given model to the data

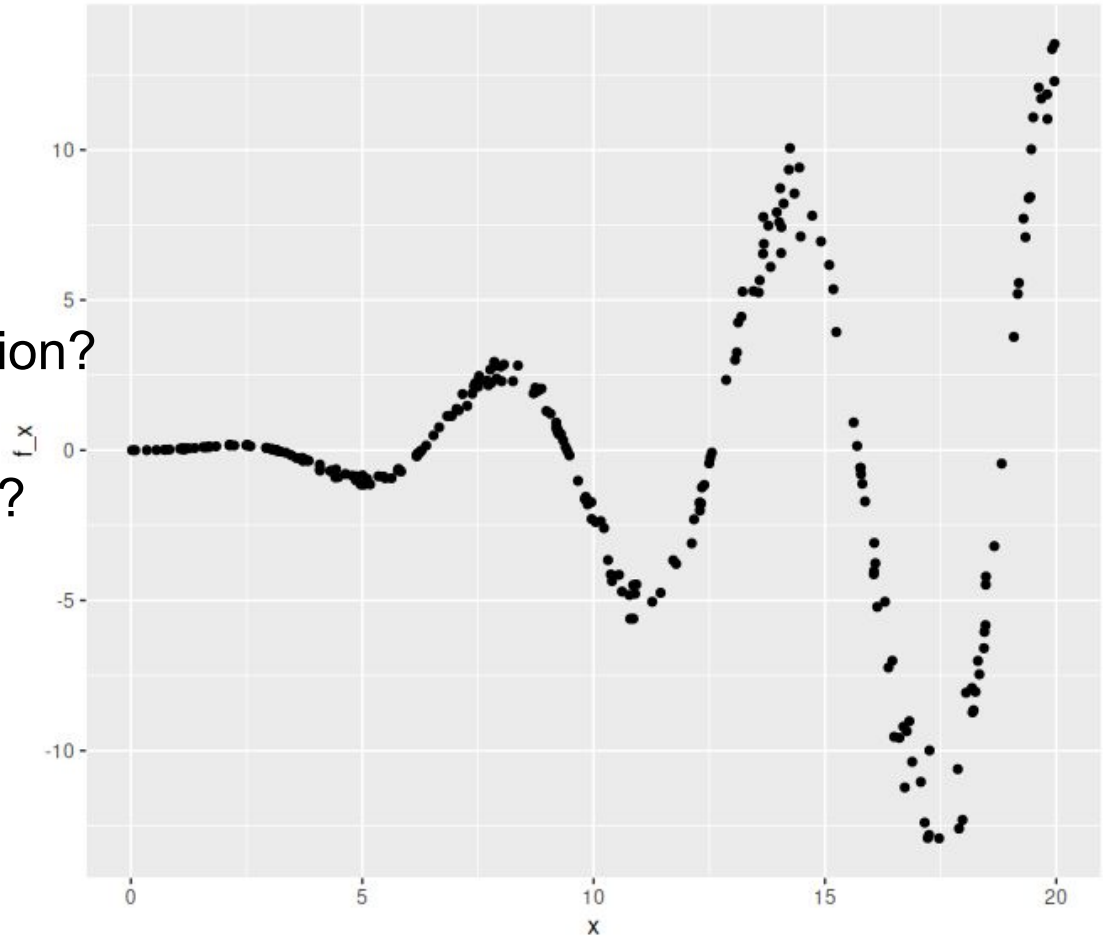
But what if we don't know
the model?

What if we evolve it?



Questions for us to answer:

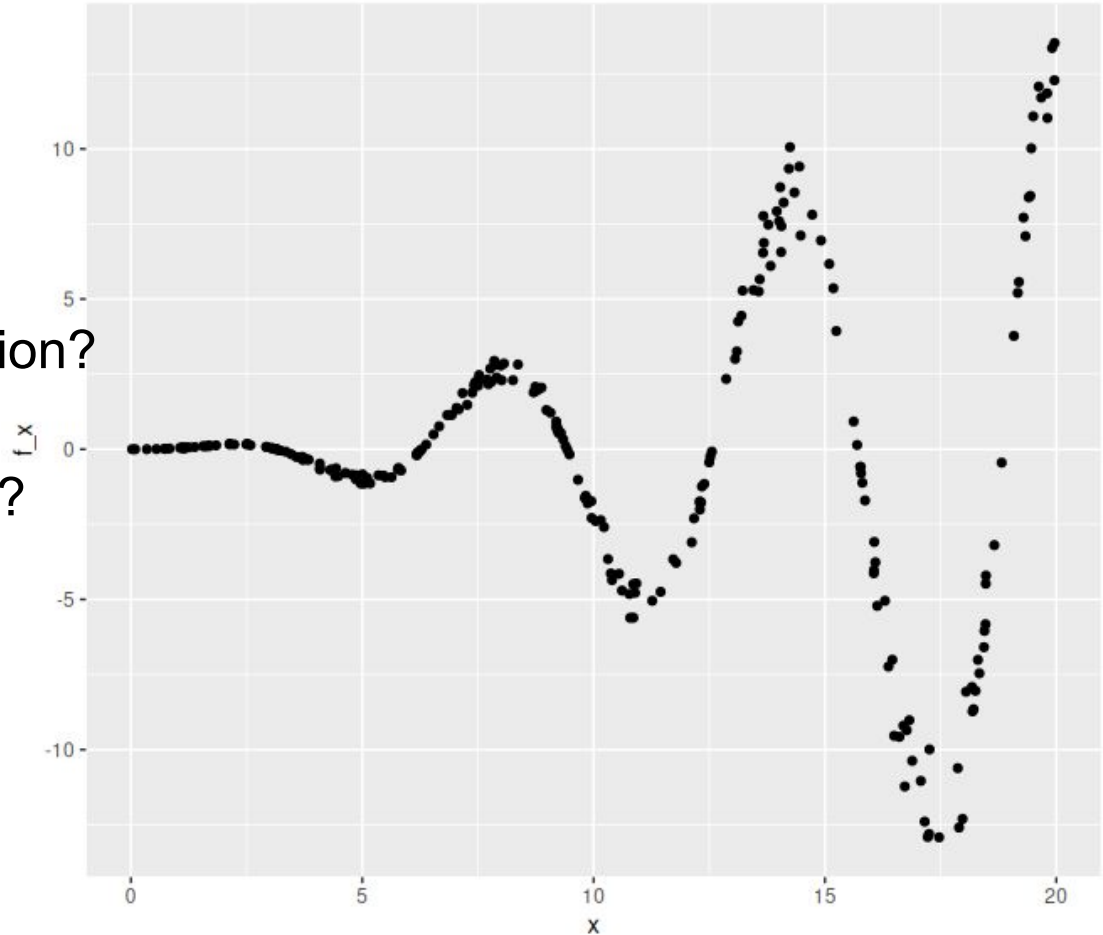
1. What is an organism / solution?
2. How big is our population?
3. How do we create variance?
4. How do we select which organisms reproduce?
5. When should we stop?



$$\frac{5 \sin(x) x^3}{120x}$$

Questions for us to answer:

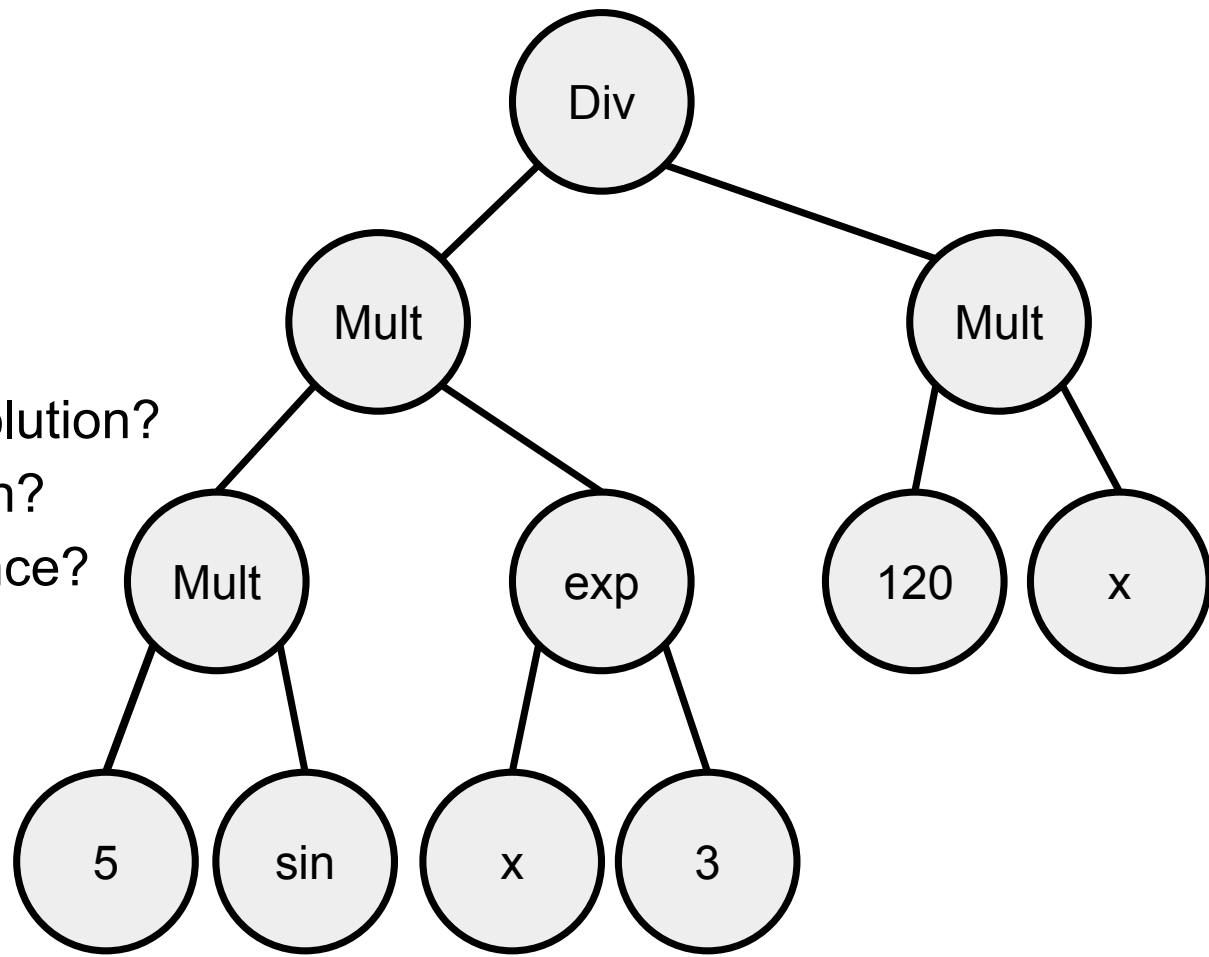
1. What is an organism / solution?
2. How big is our population?
3. How do we create variance?
4. How do we select which organisms reproduce?
5. When should we stop?



$$\frac{5 \sin(x) x^3}{120x}$$

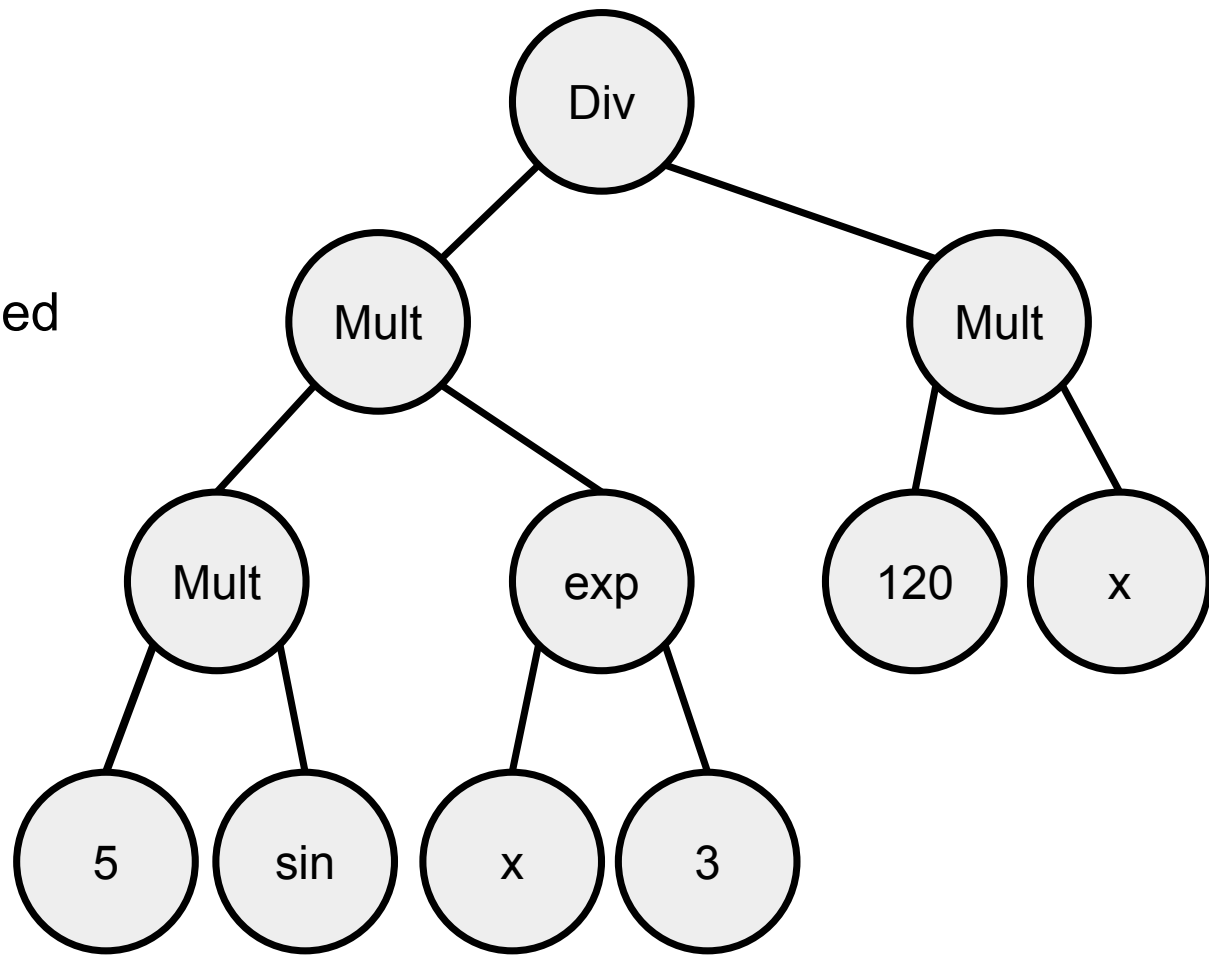
Questions for us to answer:

1. What is an organism / solution?
2. How big is our population?
3. How do we create variance?
4. How do we select which organisms reproduce?
5. When should we stop?



Symbolic regression

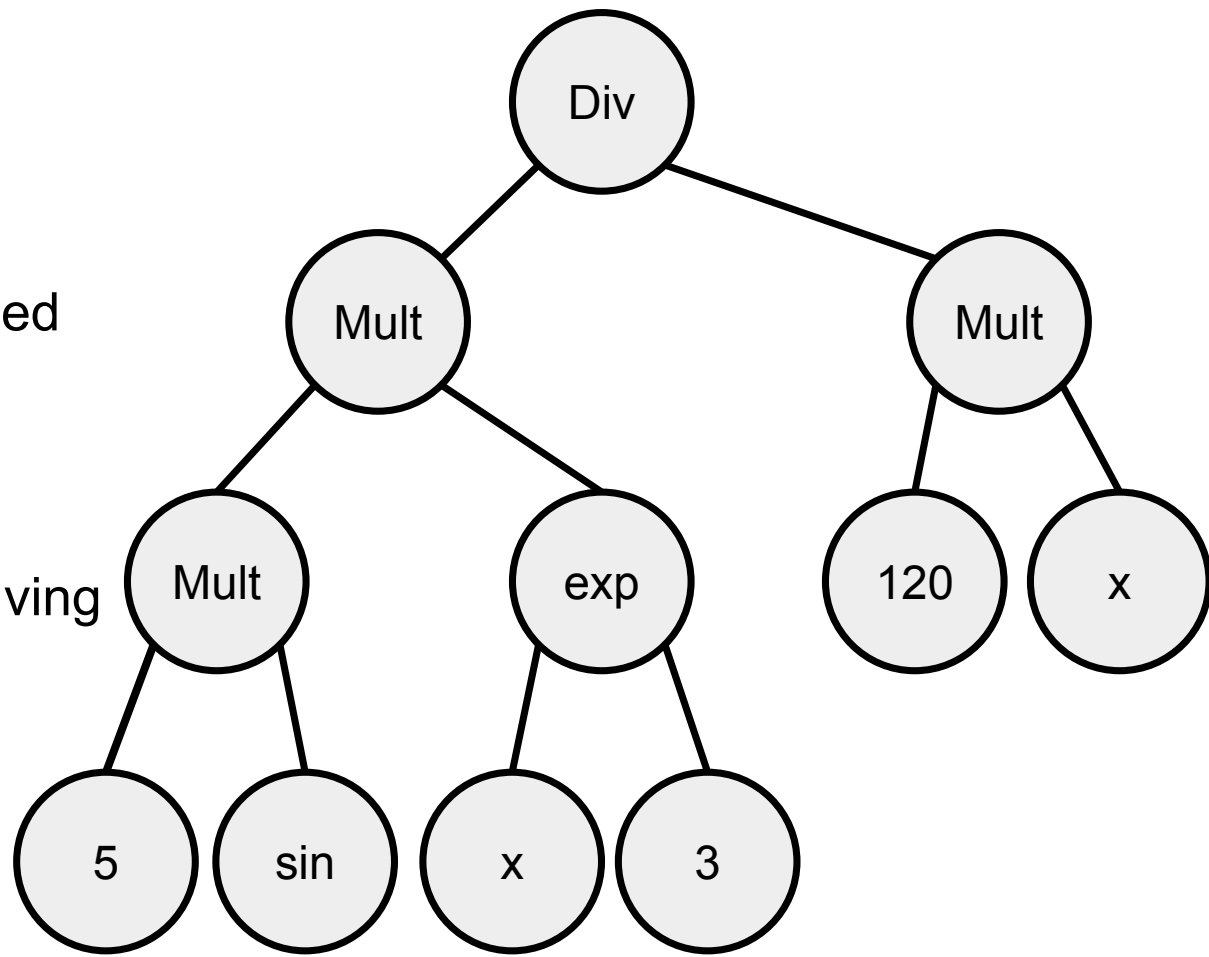
We've switched to tree-based genetic programming



Symbolic regression

We've switched to tree-based genetic programming

Genetic programming: evolving functions or code



Symbolic regression

We've switched to tree-based genetic programming

Genetic programming: evolving functions or code

We could also have used a linear GP

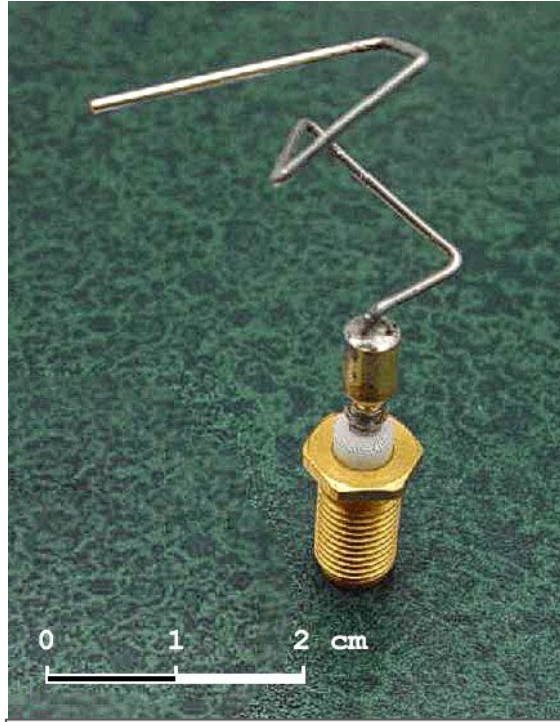
```
# reg1 is x
sin reg1 reg2 #reg2 = sin(reg1)
mult reg2 reg2 #reg2 = reg2 * 5
exp reg1 3 reg3 #reg3 = reg1 ** 3
mult reg2 reg3 reg2 #reg2=reg2*reg3
mult reg4 reg1 120 #reg4=reg1 * 120
div reg2 reg4 reg2 #reg2=reg2 / reg4
```

What has evolutionary computing actually been used for?

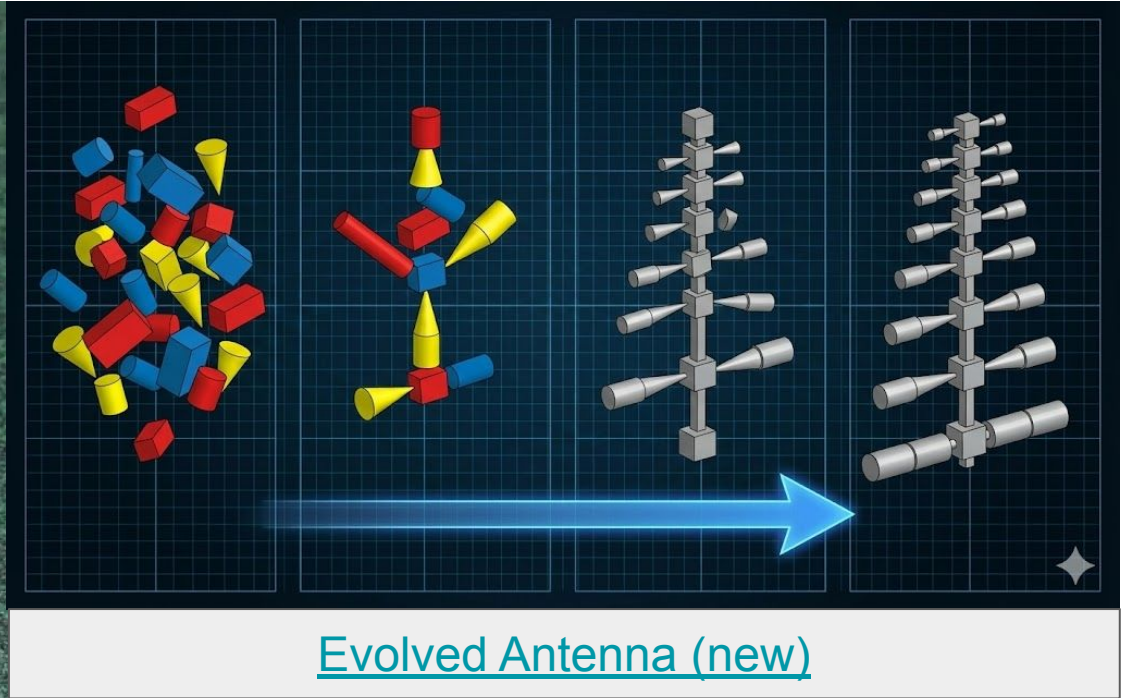


Evolved Antenna (old)

What has evolutionary computing actually been used for?

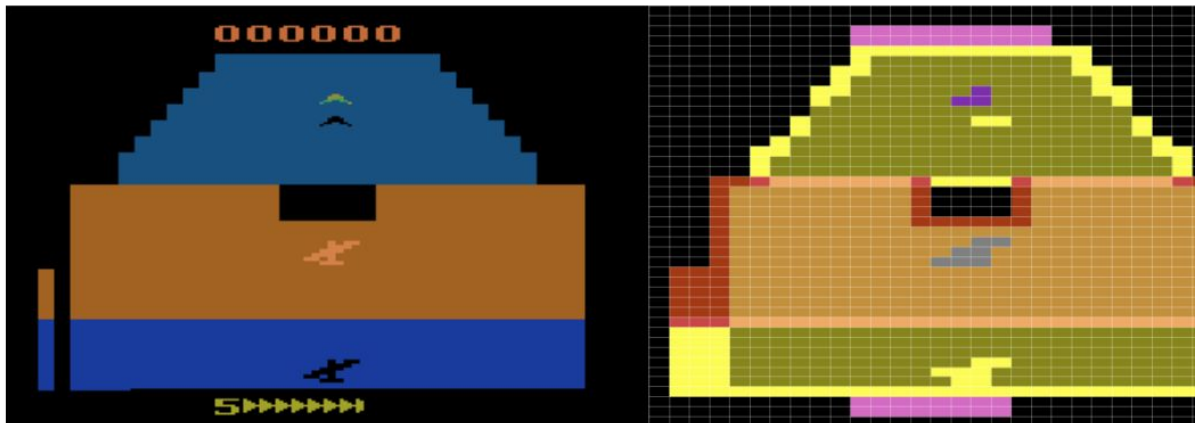


Evolved Antenna (old)



Evolved Antenna (new)

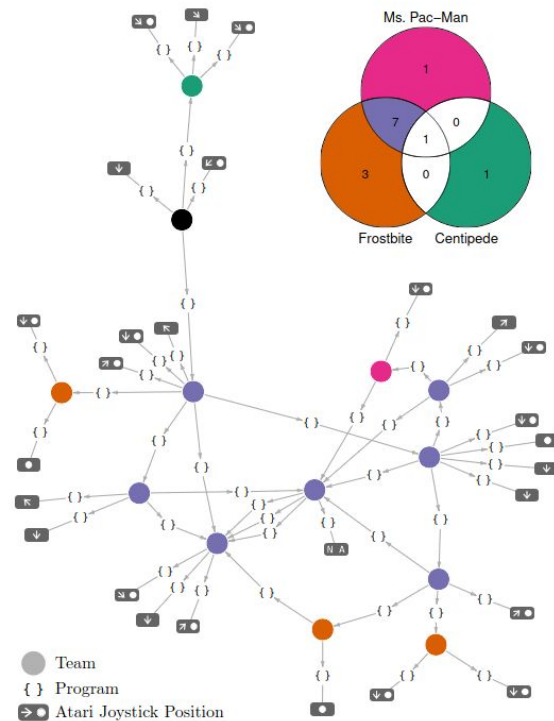
What has evolutionary computing actually been used for?



(a) Atari Screen

(b) Decimal Feature Grid

[Playing multiple atari games](#)



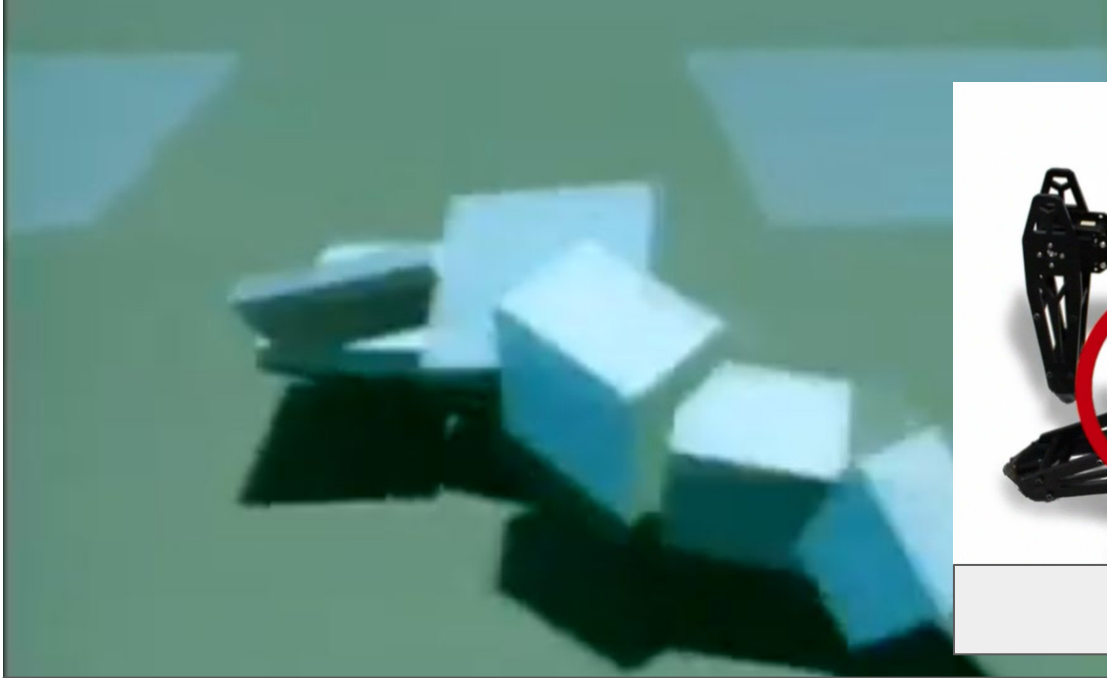
What has evolutionary computing actually been used for?

Generation 1

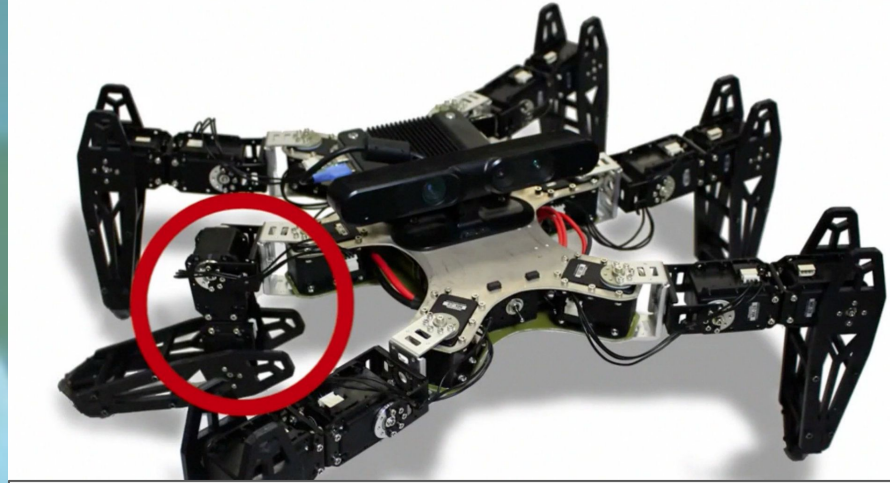


Mario

What has evolutionary computing actually been used for?

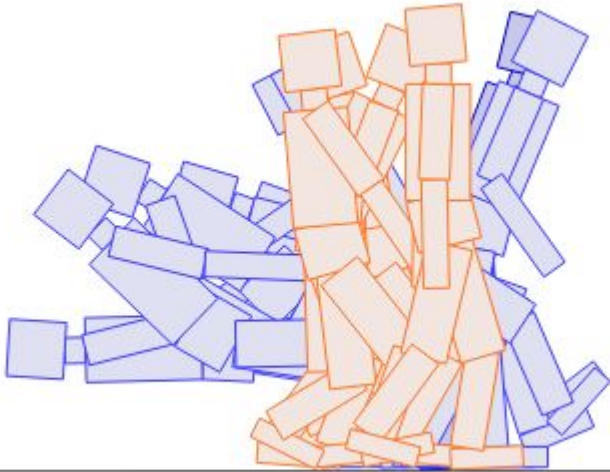


[Lots of locomotion tasks](#)

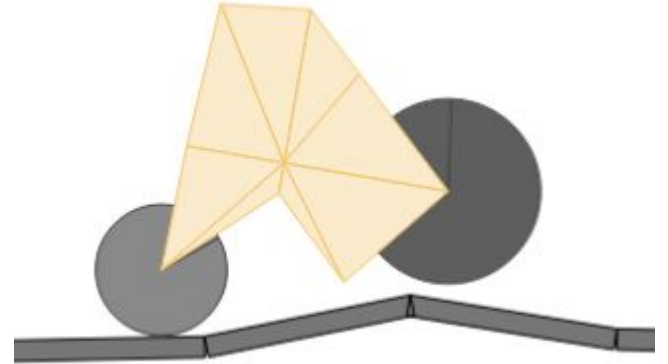


[Adaptive locomotion](#)

What has evolutionary computing actually been used for?



[Locomotion - In browser!](#)



[And cars - In browser!](#)

To paraphrase someone...

To paraphrase someone...

"Evolutionary computation is always the
second-best option for solving a problem"

How is this a whole field?

Imagine we are wanting to select parents for the next generation.

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

1. Elite selection

Always take the best
N orgs

Pros:

Cons:

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

1. Elite selection

Always take the best
N orgs

Pros:

- Simple and fast
- Guaranteed not lose best orgs

Cons:

- Little diversity

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

1. Elite selection Always take the best N orgs Pros: - Simple and fast - Guaranteed not lose best orgs Cons: - Little diversity	2. Roulette selection Make a roulette wheel based on fitness, spin! Pros: Cons:
---	---

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

1. Elite selection Always take the best N orgs Pros: - Simple and fast - Guaranteed not lose best orgs Cons: - Little diversity	2. Roulette selection Make a roulette wheel based on fitness, spin! Pros: - Every org can be selected, yet biased by fitness Cons: - Have to fine-tune fitness - Costly to run
---	--

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

1. Elite selection Always take the best N orgs Pros: - Simple and fast - Guaranteed not lose best orgs Cons: - Little diversity	2. Roulette selection Make a roulette wheel based on fitness, spin! Pros: - Every org can be selected, yet biased by fitness Cons: - Have to fine-tune fitness - Costly to run	3. Tournament selection Select N orgs at random. Best one wins! Pros: - Simple and fast - Guaranteed not lose best orgs Cons: - Little diversity
--	---	---

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

1. Elite selection Always take the best N orgs Pros: - Simple and fast - Guaranteed not lose best orgs Cons: - Little diversity	2. Roulette selection Make a roulette wheel based on fitness, spin! Pros: - Every org can be selected, yet biased by fitness Cons: - Have to fine-tune fitness - Costly to run	3. Tournament selection Select N orgs at random. Best one wins! Pros: - Tunable - Almost all orgs can be selected Cons: - No diff. Between 1 & 2 and 1&100 fitness
--	---	---

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

4. Novelty selection

Throw away fitness, select what is novel!

Pros:

Cons:

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

4. Novelty selection

Throw away fitness, select what is novel!

Pros:

- Selects for diversity
- Often simple to implement

Cons:

- May never approach objective!

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

4. Novelty selection Throw away fitness, select what is novel! Pros: - Selects for diversity - Often simple to implement Cons: - May never approach objective!	4. Age-Layered Population Structure Orgs compete with those of a similar "age" Pros: Cons:
--	--

How is this a whole field? - Example

Imagine we are wanting to select parents for the next generation.

4. Novelty selection Throw away fitness, select what is novel! Pros: - Selects for diversity - Often simple to implement Cons: - May never approach objective!	4. Age-Layered Population Structure Orgs compete with those of a similar "age" Pros: - Diversity: new orgs have time to mutate, not immediately outcompeted Cons: - Things are getting complicated! How do we determine age and brackets?
--	---

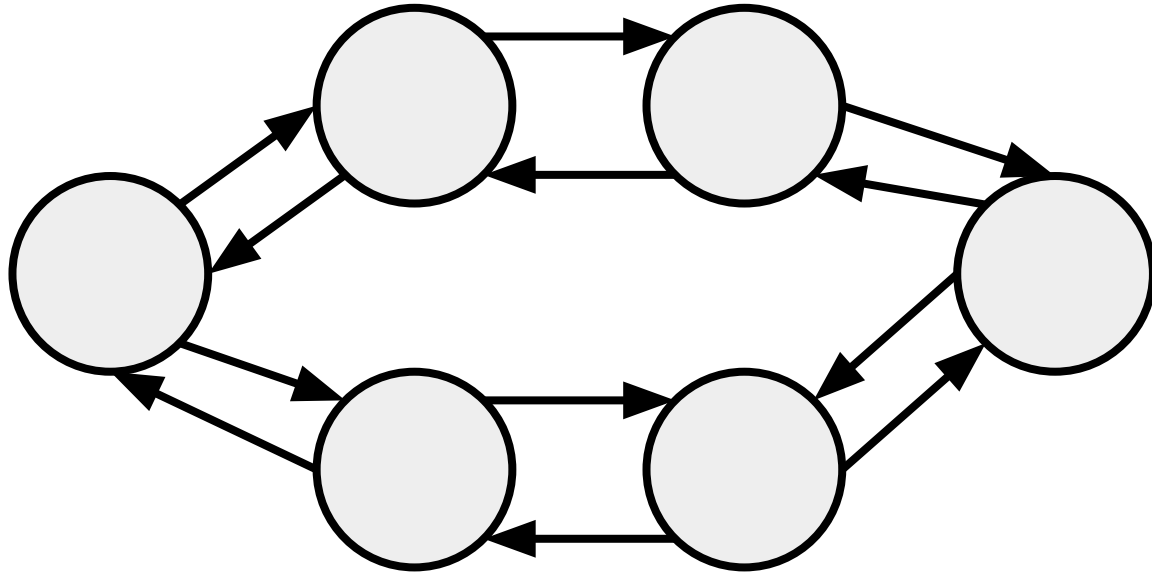
How is this a whole field? - Second example

Evolutionary algorithms often fail when diversity collapses prematurely

How is this a whole field? - Second example

Evolutionary algorithms often fail when diversity collapses prematurely

One fix? Split the population into "islands" with migration between them



What do you actually do?

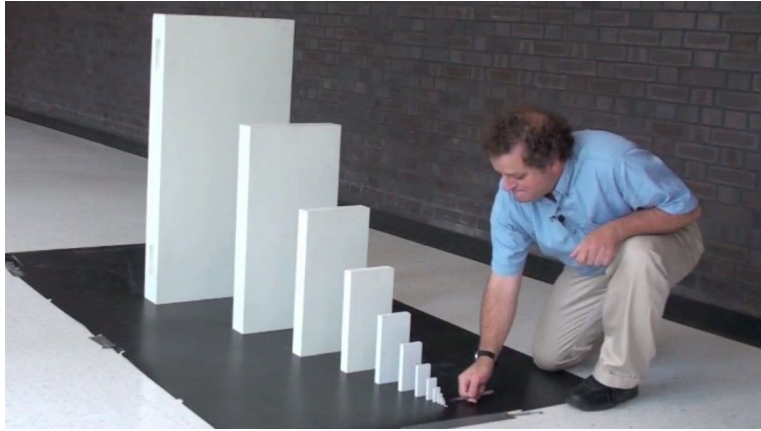
Not much of this!

I flip EC on its head:

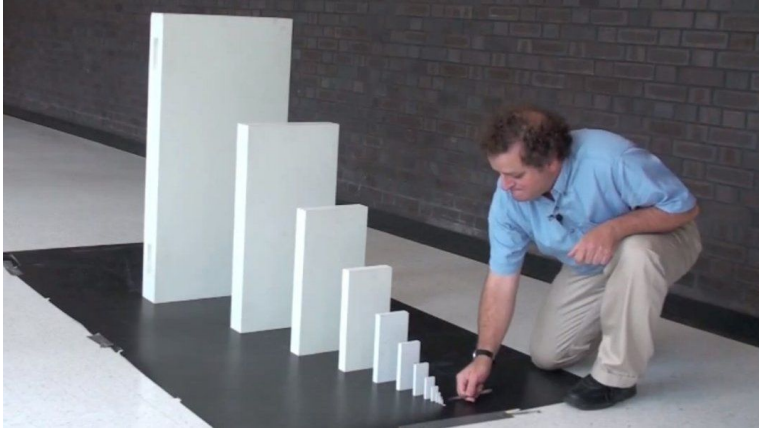
Instead of using evolution to solve computational problems...

... I use computational models to study evolution :^)

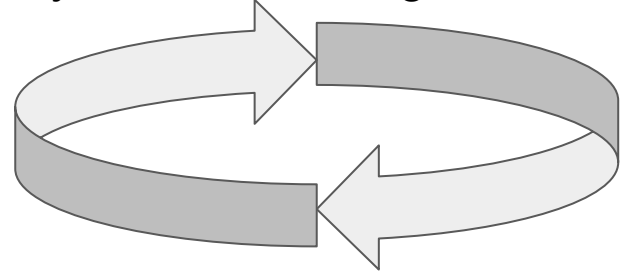
Historical contingency in evolution



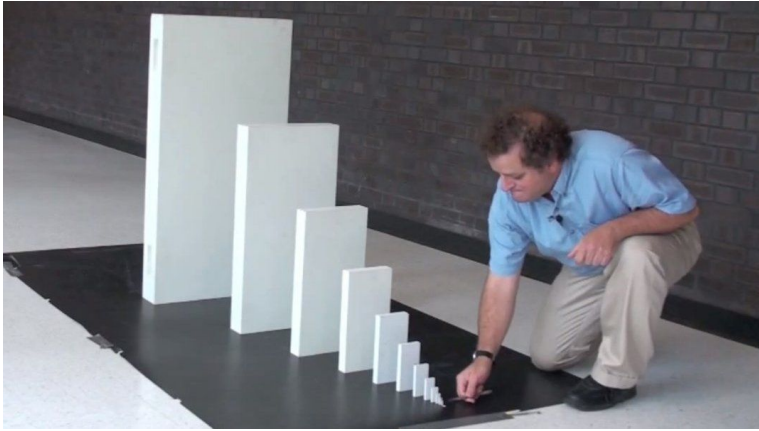
Historical contingency in evolution



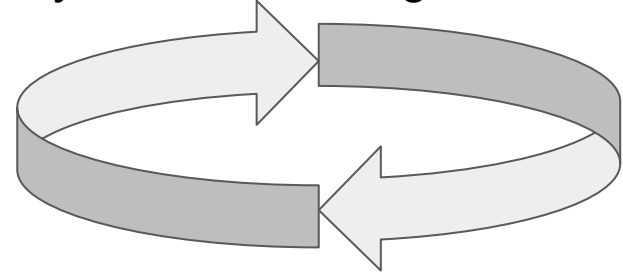
Evolvability - "how well" organisms evolve



Historical contingency in evolution



Evolvability - "how well" organisms evolve

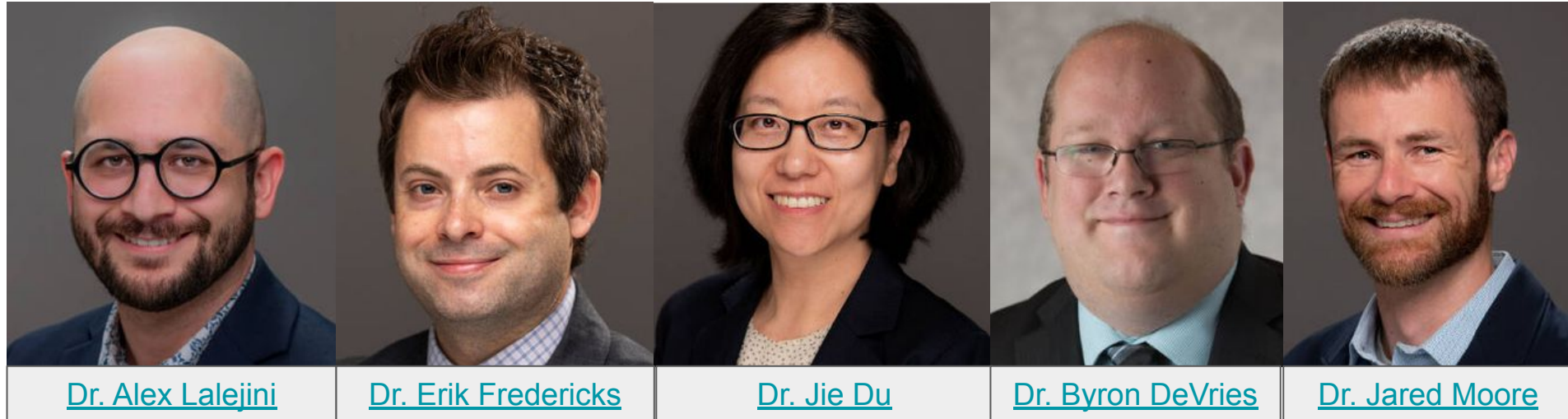


Evolutionary dynamics of spreading invasive species



[Michael Montgomery - USDA FS](#)

Other GVSU professors who use evolutionary computation





Thank you! :^)
Questions?
ferguaus@gvsu.edu